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Estimating Variation of Short-Circuit Power

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Abstract

Power systems are operated and designed to ensure proper management of faults, such as short circuits. Power generation are traditionally based on synchronous generators, these machines have a desired inherent property of supplying fault currents several times their nominal current, which makes it easier for protection relays to properly detect and clear these faults. However, the increased penetration of converter-interfaced generation also brings additional challenges both in supplying enough fault current to the power system and in quantifying the short circuit capacity throughout the system. This work tackles the latter by proposing a Python-based modular framework for continuous and automated short circuit calculation. The work aims to propose a low-effort data-driven screening method that provides indicators that may raise the need for more detailed studies using conventional commercial software. Validations of the proposed method was performed against PowerWorld and the relative error between the calculated short circuit current was less than 3 percent, for cases without converter-interfaced generation. The literature and simulations showed that synchronous machine is the main supporter of short circuit capacity, the results found that a converter-interfaced generation 20 times the size of a synchronous generator supply less than half the amount of fault current of the machine's. Although the converter interfaced generation contribution to short circuit is dependent on their control strategy, these devices' contribution are limited to ensure its components safety. Moreover, other aspects of the power system that produces changes in the short circuit capacity were also accessed and the main factors of influence found was the disconnection of synchronous generators, changes in topology and contribution of external grids. A reduced order model encoding the main factors of influence was also proposed, anticipating limited data availability and to enable faster, simpler deployment on real electrical power systems.

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Sammanfattning

Kraftsystem dimensioneras och drivs för att säkerställa en tillförlitlig hantering av fel, såsom kortslutningar. Traditionellt har elproduktion baserats på synkrongeneratorer, vilka naturligt kan leverera felströmmar som är flera gånger högre än märkströmmen. Detta underlättar för skyddssystem att upptäcka och åtgärda fel. Den ökande andelen omriktarkopplad produktion innebär dock nya utmaningar, både vad gäller tillgången på felström och möjligheten att uppskatta kortslutningseffekten i systemet. I detta arbete adresseras den senare frågan genom utvecklingen av ett modulärt, Pythonbaserat ramverk för kontinuerliga och automatiserade kortslutningsberäkningar. Syftet är att möjliggöra en datadriven och resurseffektiv screening av systemtillstånd, där indikatorer kan identifiera behovet av mer detaljerade analyser med etablerade kommersiella programvaror. Metoden har validerats mot PowerWorld och uppvisar ett relativt fel i den beräknade kortslutningsströmmen på under 3 procent i fall utan omriktarkopplad produktion. Resultaten bekräftar att synkronmaskiner utgör den dominerande källan till kortslutningseffekt. Även vid hög installerad effekt från omriktarkopplad produktion är dess bidrag till felström begränsat i jämförelse, vilket huvudsakligen beror på styrstrategier och skydd av komponenter. Vidare visar analysen att kortslutningseffekten i ett system i hög grad påverkas av förändringar i produktionsmix, nätets topologi samt kopplingen till externa nät. För att hantera situationer med begränsad datatillgång föreslås även en förenklad modell som fångar de viktigaste påverkansfaktorerna och möjliggör en snabbare tillämpning i verkliga elnät.

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Contents

Acronyms

AC	Alternate Current
CIG	Converter-Interfaced Generation
DC	Direct Current
DFIG	Doubly-fed Induction Generator
EMT	Electromagnetic Transient
ESCR	Equivalent Short Circuit Ratio
FRT	Fault Ride-Through
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
PMU	Phasor Measurement Unit
PV	Photovoltaic
SC	Short Circuit
SCADA	Supervisory Control and Data Acquisition
SCC	Short-Circuit Calculation
SCR	Short Circuit Ratio
STATCOM	STATic synchronous COMpensator
SVC	Static Var Compensator
TSO	Transmission System Operator

Chapter 1

Introduction

The Svenska kraftnät report Operational reliability in the power system [1] describes how the dynamics of the power system are changing. The entity has described this from a national level, but in principle it also applies down to local networks and 0.4 kV household customers. The report describes, among other things, how the system's inherent capabilities for fault current input decrease over time with more production connected via converter technology instead of via directly connected synchronous generators. The report also highlights how the inherent capabilities plus the supplied capabilities need to be greater than or equal to the system needs.

1.1 Objectives

A system need short-circuit power, or fault current input, in order to support enough fault current so that protection relays can detect and thereby act to maintain electrical safety. As the amount converter-interfaced generation increases on the power grid, it is expected that the short-circuit power will vary more than it has historically. This work aims to provide a framework to quantify it.

Key questions

1. How does the short-circuit power vary over time and with different types of generation?
2. Can a conceptual design be tested in an IEEE test system and implemented in SCADA?
3. How does this affect system dimensioning and safety?

1.2 Motivation and problem description

Mid-2010s Svenska kraftnät and other Nordic TSOs began to estimate the inertia in the Nordic power system [2]. This was done by establishing a calculation in the SCADA systems that summed up each individual generator's contribution to the total inertia. The calculation summed up only the generators that were phased-in to the power system, with a breaker state ON. It is pictured that a similar method could be developed for short-circuit capacity.

The priority for the result is to establish a concept sketch that can first be tested in a test system so it can be later described how the calculation can be implemented in active SCADA systems.

1.3 Scope and limitations

The method developed in this work is based on a balanced three-phase system and balanced faults, thus only the positive sequence is needed. Although these simplify the development and implementation, it represents the main limitations of this method. The proposed method requires modification (not included in this work) in order to be applied to unbalanced systems and/or asymmetrical faults. Additionally, this method is modeled considering quasi-static phasors in per unit (p.u.), therefore, high frequency phenomena cannot be captured. The electrical elements are modeled using steady-state models and the assumptions and limitation of each is discussed in their respective sections in Section 3.3.

1.4 Outline

Chapter 2 begin the formulation of the problem through a literature review on current approaches used in short-circuit calculation problems. Specifically, it investigates the methods and models found in the literature.

Chapter 3 develop the problem formulation, beginning in the theory of the problem description and then proposing the methodology to achieve this work objectives. In the final section of this chapter, the summary of the method work flow is presented.

Chapter 4 contains the simulations and discussion of the results. Since this work objective is the development of a methodology, or a framework, to make possible the estimation of short-circuit, this section begins with the validation of the results obtained from the proposed method with a commercial software, PowerWorld. Then, the response of the proposed method under different scenarios is performed to access the method resilience and coherence. As not all data required for the method might be available, a section is dedicated to quantify the error that could be found by using some assumptions. Another section is dedicated to produce results similarly as for this work inspiration, the Inertia of Nordic power system [2]. Lastly, a reduced-order model derived from the proposed method, with the minimum set of parameters possible, is presented with the goal to encourage the implementation on this method in real power systems.

Chapter 5 concludes this document and presents the relevant findings of this work.

Chapter 2

Literature Studies

Power grids were traditionally developed based on thermal (coal, oil, gas), nuclear and hydro production facilities [3, 4]. These facilities have the synchronous generator technology as one of their common factors [4]. From the power system optics, a synchronous machine have a desired inherent ability to provide inertia and short circuit capacity to the grid [1].

Although the base electricity generation worldwide still retains this synchronous generator-based characteristics, for the years to follow, significant part of the demand are set to be supplied by converter-interfaced generation (CIG), specially solar PV[3]. Nonetheless, CIG, common classification of wind and solar photovoltaics [5], could still present the desired properties of inertia [6, 7, 8] and short circuit capacity [9, 10, 11, 12, 13], but as an extended property, rather than an inherent property [1, 6].

The scenario that there is being laid is of a power system with less synchronous generator generation and, therefore, in need for innovative solutions to cope with the system needs and to understand and capture its complex dynamics. To this extend, [6] says that a grid with high penetration of CIG might even need less inertia, although the question regarding short-circuit capacity would still need to be addressed [14] in the scope of power systems protection and stability. This study aims the short circuit capacity problem.

As seen in [5, 14], the main problem regarding CIG and short-circuit capacity is that these technologies have a limited contribution to short circuit current. This limitation leads to two major issues in the power grid: fault protection [14] and voltage sags [12, 13, 15, 16, 17] and consequently their Fault Ride-Through (FRT) requirements at grid codes [18, 19].

In the context of this work, the fault protection issue relates to the condition where insufficient fault current makes a traditional over-current protection device unable to detect the fault, and consequently, unable to properly actuate on it. In the work of [14] it is discussed the scenarios for this problem. Voltage sags, which are a consequence of the inability of the system to maintain steady voltages on a specified level after an event. The latter falls into the scope of Voltage Stability and Converter-driven Stability, as discussed in [5]. Things considered, the assessment of these issues begins with the short circuit capacity quantification.

A continuous quantification of the ”system strength” through its short-circuit capacity (S_{SC}), as proposed on this work, might help identify beforehand vulnerable areas [5] so that detailed studies could be conducted [4]. Converter-driven stability became a dedicated power system stability category [5] and Short Circuit Ratio (SCR) is a well known metric (Equation 2.1) used to define system strength. However, as CIG becomes more relevant in the grid, this metric is being replaced by others [4] such as the Equivalent Short Circuit Ratio (ESCR) as in Equation 2.2. These metrics quantify how sensitive a point in the network is to the local production. Higher values, also known as "strong/stiff grid", are desirable and means that that point of the system is not highly affected by local production. Intuitively, this shows the system stability dependence on the local production.

$$SCR = \frac{S_{SC}}{P_{con}} \tag{2.1}$$

S_{SC} is the short circuit power at the connection point and the object of interest in the Short-Circuit Calculation and P_{con} is the converter facility active power in the same connection point.

$$ESCR_i = \frac{S_{SC}}{P_i + \sum \frac{\Delta V_j}{\Delta V_i} P_j} \tag{2.2}$$

P_i is the converter facility active power and ΔV_i the voltage change in the bus under study. P_j and ΔV_j is the correspondent values for other converter-connected buses. $\frac{\Delta V_j}{\Delta V_i}$ quantifies the change of voltage in bus j cause by changes in the studied bus i. The difference between SCR and ESCR is that the latter consider the interaction between all CIG in the network.

To determine S_{SC} in Short-Circuit Calculation in large-scale power systems, matrix notation if often used [20, 21] because an arbitrary N-bus can be conveniently represented in a computer program [22]. As extension, three-phase AC systems the calculation of the current values resulting from balanced and unbalanced short circuits is simplified by the use of symmetrical components [21, 23, 24].

Traditionally, in SCC, the faulted system state is obtained by superposition of the calculated Δ-circuit state and the known pre-fault state (e.g. power flow) [25], but as CIG are included in the system, traditional modeling and calculations for SCC are not sufficient [25, 26] since they are decoupled from the system by power electronics. There are methods in the literature that deals with the SCC including CIG such as: the *equivalent voltage source* method [23, 27], the *load-flow-based* method [28], the *modified superposition* method [25] and the most common *iterative* method [29, 30, 31, 32]. In the previous works mentioned, only the ones based on IEC 60909 [23, 27] explicitly handles the DC component.

2.1 Superposition and iterative method

One of the main component to the Short-Circuit Calculation is the nodal admittance matrix (Y_{bus}), which represents the network topology and its elements. For short-circuit analysis on transmission systems, all elements are individually included in order to avoid loss of accuracy [20]. In reality, actual data for some equipments might not be known, so typical values are either assumed [20] or estimated using voltage and current measurements [33, 34].

The system can be represented through its nodal equations, Equation 2.3, where E is the vector of bus voltages and I the current injections.

$$Y_{bus}E = I \tag{2.3}$$

$$Z_{bus}I = E \tag{2.4}$$

$$I_{SC} = \frac{V_{f,n}}{Z_{nn}} \tag{2.5}$$

At the Δ-circuit state, all generators internal voltage sources are shorted and the only voltage source remaining is at fault location [21]. From Equation 2.4, since the Δ-circuit now contains only one source, the current vector contains only one non-zero entry, at the fault location, thus the linear system of equations solving for the fault current is reduced to Equation 2.5. Mapping Equation 2.4 to Equation 2.5, the fault current $I_n = -I_{SC}$ at the $n-th$ bus is calculated by the equivalent voltage at fault location $E_n = -V_{f,n}$ and the equivalent bus impedance Z_{nn}, this is the traditional superposition method.

When introducing CIG to the system, I_{SC} from cannot be directly calculated using Equation 2.5 because the circuit is no longer linear since the converter current injection is a function of its own terminal voltage. Here enters the different Short-Circuit Calculation methods as introduced previously.

Equation 2.3 and Equation 2.4 implies a balanced quasi-sinusoidal system. Although the previous implication might not withstand for detailed stability studies [5] and CIG dynamic response [5, 9], the study of the faults in steady state is convenient because it is predictable for various control objectives [9].

To accurate capture CIG short-circuit dynamic in sub-transient and transient timescale (millisecond resolution) would require the detailed information of the hardware nature and its internal control [9], which is beyond the objective of this work and redundant to the forthcoming detailed studies. For the sake of comparison, conventional SCADA systems usually sample data at the order of 1 Hz [35], electricity price markets have a Market Time Unit (MTU) of 15 minutes [36] and the similar-to-this-work higher-level assessment of inertia, a resolution of 15 minutes as well [2]. The next section extends the discussion relating timescale and accuracy.

After the previous remarking, we start to discuss the elements that compose the Short-Circuit Calculation.

2.2 Production facilities

The magnitude of short-circuit current is primarily determined by the amount of connected generation plant (production facilities) and network topology [20]. International standards as IEC 60909:0 [23] proposes conservative results [28], and current literature propose models to account the different behavior of Converter-interfaced generation throughout the different time scales [9]. The element model is also dependent on the method chosen for Short-Circuit Calculation.

Power-generating modules classification is defined in the European Requirement for Generators [19], however, to maintain the classification found in the literature [9, 25, 26], the production facilities in this work are defined as follows:

Type A: synchronous machine directly connected to the system, e.g. small and medium scale hydro turbines, gas turbines, etc.

Type B: induction machine directly connected to the system, e.g. microscale hydro turbines, wind turbines with fixed speed, internal combustion engines, etc.

Type C: Double-fed induction machine (DFIM), wind turbines with partially variable speed.

Type D: Inverter-interfaced generation (IIG), e.g. photovoltaics, wind turbines with fully variable speed, energy storages, plug in electric vehicles, fuel cells, etc.

Overall, from a Short-Circuit Calculation perspective, Type A and Type B does not require modification on the superposition method whereas Type C and Type D does [25].

Type A in fault analysis are modeled as an impedance $(R + jX)$ in series of a voltage source [20, 21]. It captures the effect of the magnetic flux caused by the short-circuit armature current. During the fault, the reactance (X) increases over time and its physical explanation [21] is that the magnetic flux caused by the short-circuit armature currents is initially forced to flow through high-reluctance paths that do not link the field winding or damper circuits of the machine. IEC 60909 [23] applies correction factor on the generator impedance in their method.

Type B Asynchronous Generators includes but are not limited to Type-1 and Type-2 Wind turbines. It is modeled similarly to the synchronous generator, a voltage source behind an impedance. The impedance is calculated using stator leakage reactance (X_s), the magnetizing reactance (X_m), the rotor leakage resistance (X_2), the stator resistance (R_s) and the rotor resistance (R_2) [37, 38]. On the other hand, IEC 60909 method rather uses nameplate values, e.g. voltage, current, apparent power and locked rotor current to compute the equivalent impedance [23]. As extension, from a system perspective, is of interest the contribution of a wind power plant rather than a single generator. Starting from the individual machines, the power plant impedance (single-source equivalent) will be the parallel of those machines' impedance. All cable impedances in the collector circuit can be neglected without significantly affecting the accuracy of the equivalent model for faults external to the wind plant [37].

Type C is a category for partly electronically-couple connection, i.e., the stator is directly grid-connected, but the rotor has a power electronics connection. This is characteristics of a Type-3 Wind Turbine. A remarked characteristic of these machines is the presence of a crowbar. The performance of these machines might be divided into two states: In the crowbarred state, the fault behavior is defined by the flux equations of the physical machine, similar to Type B. When the crowbar is not engaged, however, the machine operates according to its control design [37]. A highly accurate modeling of a Type C is only achieved by Electromagnetic Transient Programs (EMT), However EMT programs are not a practical means of performing Short-Circuit Calculation in large networks [37], such as in this work. Thus, a trade-off between accuracy and practicality must be made.

Solutions in the literature divide the operation Type C machine into different timescale as with some consideration factors regarding the hardware and control dynamics [9], consider only the machine physics in the non-crowbarred state and neglect the nonlinear (power electronics control) state [37], using empirical/manufacturer data within the IEC 60909 method [23], an additional voltage source to the Type B model representing the rotor [38], fault ride-through grid requirements [39] and a voltage-current mapping [40] and as a controlled current source [9, 25]. Then [41, 42, 43] discuss the single-source equivalent for this type of machines.

Type D uses a converter for grid connection, as in Solar PV and Type-4 Wind Turbine. The main structure is composed of outer/inner feedback control loops, Phase Locked Loop, pulse width modulation and filter [9]. The contribution to short-circuit current is limited to rated or little above rated [38]. At subtransient stage the short-circuit characteristics of these sources comes mainly from the filter, the controller in the transient stage and afterward, in a quasi-steady state, determined by current limitation and the specific control strategy for the duration of the fault ride-through [9] as in grid codes [18, 19].

Type D presents the same modeling accuracy challenges as Type C [37], the nonlinear operation decurrent of the power electronics control. However, most literature approaches the modeling similarly: a controlled current source [9, 11, 23, 25, 26, 37, 38, 42]. Other approaches use an iterative method using a voltage-behind-impedance model to output a certain current [44]. Similarly for Type B and Type C, [41, 42, 43] discuss the single-source equivalent.

Further power grid elements are discusses in the next section.

2.3 Additional grid elements

Uncontrolled shunt capacitor and reactor banks acts as passive network elements, i.e., incorporated in the Y_{bus}, sometimes neglected [21]. STATCOM have a similar behavior of Type D facilities [37] and a further discussed in [45, 46, 47]. Synchronous condensers present similar behavior to Synchronous generators (Type A) [23, 45]. Additionally, within the IEC 60909 method, shunt admittances of non-rotating loads are neglected [23]. Short-Circuit Calculation commonly does not include loads because fault current from generators are several times larger than the load current, therefore their omission does not significantly reduce the accuracy of the calculations [48].
The main challenges of this field of research, Short-Circuit Calculation, is its limitation regarding a direct application for protection designs [9]. Although some branches of research are acting on this [49, 50, 51], network calculation for the subtransient and transient stages are still absent [9] due to the wide range of control strategies for CIG [26].

Lastly, several uncertainties might arise during a Short-Circuit Calculation: tolerance on equipment manufacturing, on the available data, the employed method [20] and statistical fluctuation from field measurements [52].

This work objectives a higher-level assessment of the short-circuit level and its evolution over time, which serves as a screening method to access the need for more detailed studies [4]. The method develop in this work will be based on current literature and industry standards as it will also serve as an evaluation of the requirements for the deployment of short-circuit estimators in real power systems.

Chapter 3

Method

This chapter contains the problem formulation with the definition of the main variables of interest and input data. It covers the solution method, the models and the methodology implementation workflow.

3.1 Problem Formulation

For short circuit calculation, the interest is to determine the currents and voltages of the nodes (buses) in case of faults. The development of this work lies on two main premises: the network is balanced, and the fault is a symmetrical short circuit. For a transmission system analysis, the first premise is reasonable [53] due to its inherent balanced design regarding phase loading and line transposition [23], therefore, only the positive sequence network needs to be analyzed. The second premise however, it is a limitation in this work results towards unsymmetrical faults. Three-phase faults without ground can be adjusted for, performing a correction on the fault impedance equivalent [54], although, while considering the order of dimension between the system equivalent impedance in comparison to the fault impedance, the results for symmetrical short circuits with and without ground could be approximated as similar without significant loss of accuracy, in case of negligible fault impedance and grounding impedance.

The method that will be presented hereafter uses the per unit (p.u.) system to represent the electrical circuit. Although it is a correct representation when the base values are coherent, base values on real power system are not always coherent [23] and so a further limitation of this work results. The transformers phase shift are omitted, as the object of interest is in the magnitude of voltages and currents [21]. Unless stated otherwise, $MVA_{base} = 100\,MVA$ and the voltage base is according to the transformers' ratio.

The words *node* and *bus* will be used interchangeably in this work. Consider a network with N nodes (buses). Figure 3.1 shows an arbitrary node $'m'$ of the network including some of the modeled elements that will be presented in a forthcoming Section 3.3. Not including all possible elements do not cause the loss of generality of the method as one might have noticed. Additionally, a node can have an arbitrary number of neighboring nodes, meaning that the method is suited for both mashed and radial networks.

The proposed method for estimating the short circuit current in every node of a network starts from the Kirchhoff's Current Law (KCL) for a node m.

$$I_m = \sum_i I_{m,i} = 0 \tag{3.1}$$

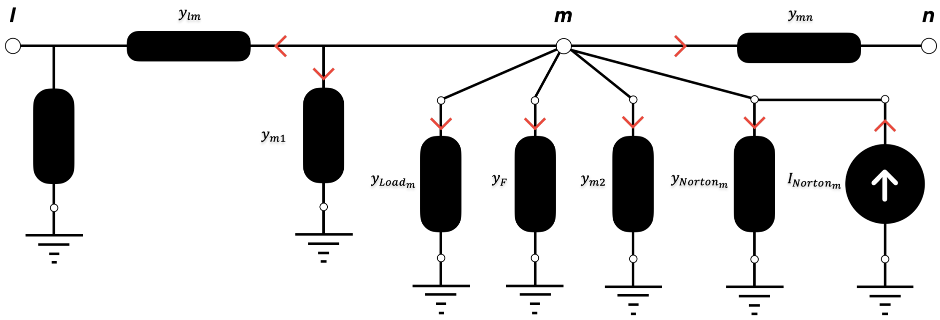


Figure 3.1: A section of an example network, showing the electrical elements and their current direction convention. All currents are set to leave the node, except for the current sources.

Where i is each current contribution to node m.

Referencing Figure 3.1, expanding each term of the summation and rearranging it, yields

$$(y_{lm} + y_{m1} + y_{m2} + y_{mn} + y_{Load_m} + y_{Norton_m} + y_F)\, V_m - y_{lm}V_l - y_{mn}V_n = I_{Norton_m} \tag{3.2}$$

The terms $y_{lm} + y_{m1} + y_{m2} + y_{mn} = Y_{mm}$ will be referenced hereafter as the element of the passive admittance matrix $Y_{passive}$ as they can be assumed constant for a fixed network topology.

Since KCL must hold for all nodes, the previous Equation 3.2 can be generalized as

$$[Y_{SCC}]_{N\times N} \cdot [V_{fault}]_{N\times 1} = [I_{injection}]_{N\times 1} \tag{3.3}$$

where $[Y_{SCC}]_{N\times N}$, is the short circuit (SCC) impedance matrix for a fault in the $m-th$ node, defined as

$$\begin{bmatrix} Y_{11} + y_{active_1} & \dots & Y_{1m} & \dots & Y_{1N} \\ \vdots & \ddots & \vdots & & \vdots \\ Y_{m1} & \dots & Y_{mm} + y_F + y_{active_m} & \dots & Y_{mN} \\ \vdots & & \vdots & \ddots & \vdots \\ Y_{N1} & \dots & Y_{Nm} & \dots & Y_{NN} + y_{active_N} \end{bmatrix} \tag{3.4}$$

Y_{active} represent the Norton equivalent of the active elements such as Loads, Synchronous Generators/Condensers, Asynchronous Machines and Full Converters. The matrix from Equation 3.4 is symmetrical, and the off-diagonal elements have their definition similarly as in a conventional nodal admittance matrix.

From Equation 3.4 is seen that the short circuit calculation method only alters the diagonal elements of the final admittance matrix, and in addition to the matrix sparsity characteristics, the following notation provides a practical implementation for the Y_{SCC}.

$$[Y_{SCC}] = [Y_{passive}] + [Y_F] + [Y_{active}]$$

$$[Y_{SCC}] = [Y_{passive}] + [Y_F] + [Y_{Load}] + [Y_{SG}] + [Y_{Converter}] \tag{3.5}$$

Since $Y_{passive}$ is only needed to be built once and the remaining matrices have only diagonal elements, the resulting matrix Y_{SCC} benefits from flexibility and computational performance. All matrices from Equation 3.5 have dimension $N \times N$. Table 3.1 provides their definition.

Table 3.1: Definition of matrices used in the short circuit calculation.

Variable	Definition
Y_{SCC}	Matrix used in short circuit calculations
$Y_{passive}$	Composed by transmission lines, transformers and uncontrolled shunt elements
Y_F	The fault admittance, is zero except in the fault location
Y_{active}	The aggregation of the following 4 matrices
Y_{Load}	Contains impedances of all loads
Y_{SG}	Contains impedances of all synchronous generators
$Y_{Converter}$	Contains impedances of all converters
$Y_{External\,grid}$	Contains impedances of all external grids

The remaining matrices $[V_{fault}]$ and $[I_{injection}]$ from Equation 3.3 are the voltage in each node of the network during the fault condition and the contribution of the current sources in the nodes, respectively.

The solution of Equation 3.3 is the nodes (buses) voltages. Once the voltage (V_{fault}) and the inserted fault impedance (y_F) at bus m is known, the short circuit current (I_{SC}) is readily calculated through Ohm's law.

$$I_{SC,m} = \frac{V_{fault,m}}{z_F} \tag{3.6}$$

3.2 Solution method

This section presents the main algorithm to solve the nodal fault voltage from Equation 3.3.

When using only linear elements, e.g. a network where the sources have only Synchronous Machines, the problem is the traditional short circuit calculation method, and the solution can be found by solving the linear system of equations [21, 20, 55, 22, 56].

However, the proposition of this work is a general solution using the Newton-Raphson method, valid for both linear and nonlinear systems, which makes it suitable for analyzing short circuits in modern power system, i.e. with converter-interfaced renewable generation and other nonlinear elements.

3.2.1 Short circuit calculation

In the general solution, an iterative method is needed to solve Equation 3.3 since some current sources are voltage dependent (nonlinearity). A general solution with an iterative method is preferred since it can handle both cases, and with a one-iteration convergence for linear cases. The solution algorithm uses the Newton-Raphson method on Equation 3.3 to solve for

$$f(V_{fault}) = [Y_{SCC}] \cdot [V_{fault}] - [I_{injection}] = 0 \tag{3.7}$$

The algorithm framed to the Short-circuit calculation is presented in Algorithm 1.

Algorithm 1 Short Circuit Calculation Algorithm with Newton-Raphson

Initialize $V_{fault} = V_{pre-fault}$
Compose Y_{SCC}
Initialize empty I_{inj}
Add I_{SG} to I_{inj} ▷ Since voltage invariant
while $i < max$ **do**
    Set $I_{Load}, I_{Converter}$ to zero
    Compute $I_{Load}, I_{Converter}$ with V_{fault}
    Add $I_{Load}, I_{Converter}$ to I_{inj}
    Compute $f = Y_{SCC} V_{fault} - I_{inj}$
    if $|f| \leq \epsilon$ **then**
        break
    end if
    Compute Jacobian $J = \frac{\partial I}{\partial V}$
    Solve $J \cdot dV = -f$ for dV
    $V_{fault} = V_{fault} + dV$
end while
return V_{fault}

One might notice that the pre-fault condition is of importance to calculate the faulted condition. The pre-fault condition is used to initialize the model of the active elements, e.g. Synchronous generators/condensers, Asynchronous machines, converters, loads. In this work the pre-fault condition is determined by a power flow solution, however, on a real power system, these are often available as measurements in the supervisory system [22].

An advantage of the method presented in this work is that it does not assume pre-faulted states, instead, it estimates realistic condition based on the actual network configuration. This joint implementation provides flexibility for test and analysis and, also eliminates the need of additional third party software and integrations. The power flow solution algorithm is presented next.

3.2.2 Pre-fault calculation

The power flow is an artificial problem [22], and in the scope of this work serves only the purpose of emulating the measurements taken from a real supervisory system. Alternatively, another approach is to assume fixed pre-fault voltage values, for this, a dedicated section in the results (Section 4.4.4) quantifies the error in this assumption.

Pre-fault states are estimated by a power flow solution using the Newton-Raphson method. The algorithm is commonly found in power systems literature [21, 20, 24], but will be presented here in a simplified manner, for the sake of continuity.

The core idea of the power flow problem is to find the voltages magnitude and angle for each node (bus) of a system given it power injection. The electrical governing equations are

$$P_m = V_m \sum_{i=1}^{N} V_i \left[G_{m,i} \cos\theta_{mi} + B_{m,i} \sin\theta_{mi}\right] = P_{gen,m} - P_{load,m} \tag{3.8}$$

$$Q_m = V_m \sum_{i=1}^{N} V_i \left[G_{m,i} \sin\theta_{mi} - B_{m,i} \cos\theta_{mi}\right] = Q_{gen,m} - Q_{load,m} \tag{3.9}$$

Where, for a bus m, P_m/Q_m is the net power injection, V_m is the voltage magnitude, V_i is the voltage magnitude of adjacent buses, $G_{m,i}/B_{m,i}$ is elements of the nodal admittance matrix ($Y_{passive}$, covered in a section to follow) and $\theta_{mi} = \theta_m - \theta_i$ is the angle difference between bus m and its adjacent. N it is the number of nodes (buses) in the network. $P/Q_{gen,m}$, $P/Q_{load,m}$ are the power injection of generation and load into the bus.

The functions to be solved by the Newton-Raphson method are

$$P_{inj} - P(|V|,\theta) = \begin{bmatrix} P_1 \\ \vdots \\ P_m \\ \vdots \\ P_N \end{bmatrix} - \begin{bmatrix} P_1(|V|,\theta) \\ \vdots \\ P_m(|V|,\theta) \\ \vdots \\ P_N(|V|,\theta) \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tag{3.10}$$

$$Q_{inj} - Q(|V|,\theta) = \begin{bmatrix} Q_1 \\ \vdots \\ Q_m \\ \vdots \\ Q_N \end{bmatrix} - \begin{bmatrix} Q_1(|V|,\theta) \\ \vdots \\ Q_m(|V|,\theta) \\ \vdots \\ Q_N(|V|,\theta) \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tag{3.11}$$

$P/Q_m(|V|,\theta)$ are the net power injection from Equation 3.8 and Equation 3.9. Note that the $|V|,\theta$ are column vectors of dimension $N \times 1$, meaning that the net injection on each bus is dependent of every other bus adjacent to it.

Each bus then have four variables $|V|, \theta, P_{inj}, Q_{inj}$. There are $4N$ variables for $2N$ equations (Equation 3.8, Equation 3.9). This is an underdetermined system with infinite many solutions, so two of the four variables must be set for each bus to get a unique solution. These are denominated as PQ (set P and Q), PV (set P and V) and Slack (set V and θ) buses. Full converter sources are modeled here as constant P and Q injections, i.e. a load with inverted sign.

Define the Jacobian as

$$J = \begin{bmatrix} \frac{\partial P}{\partial \theta} & \frac{\partial P}{\partial V} \\ \frac{\partial Q}{\partial \theta} & \frac{\partial P}{\partial V} \end{bmatrix} \tag{3.12}$$

Algorithm 2 presents the solution algorithm.

Algorithm 2 Power Flow Algorithm with Newton-Raphson

```
Initialize $P_{inj}, Q_{inj}$
Initial guess $x = (|V|, \theta)$
while $i < max$ do
    Calculate $P(x), Q(x)$
    Compute mismatch $P_{inj} - P(x)$ and $Q_{inj} - Q(x)$
    if $|mismatch| < \epsilon$ then
        break
    end if
    Compute Jacobian
    Solve $J \cdot dx = -mismatch$ for $dx$
    $x = x + dx$
    Reset voltage for PV buses
    Enforce generator reactive limits
end while
return $x$
```

Once the buses voltages are known, what is left is to compute the complex power $S = P + jQ$ for the active elements, as it is input data for their model.

The power flow solution provides a lumped value for all elements connected to the bus, without distinction of individual contributions (power dispatch). The dispatch of loads and full converters is trivial since those have defined PQ values. Generators have defined P value but not Q. For this method, the reactive power dispatch is made through the inverse of the generator impedance, i.e. lower impedance generators will have larger reactive power.

The next section presents the admittance matrices definitions, construction and clarify the difference between the matrix used in short circuit calculation (Y_{SCC}) and power flow ($Y_{passive}$).

3.2.3 Bus admittance matrix

The bus admittance matrix represents the topology of a power system. The method presented in this work makes use of two distinguish but related admittance matrices.

The first one, the passive admittance matrix ($Y_{passive}$) contains, as the name suggest, passive elements. These are elements that are represented by their impedance, such as transmission lines, transformers and uncontrolled reactive shunt devices. In the context of power flow, this specific matrix is used since all other elements (active elements) are represented as power injection sources.

In the short circuit context, active elements are modeled as a Norton equivalent, therefore, their impedance will be part of the system equivalent. Active elements' impedance also influences the short circuit level and dynamics as highlighted previously on the literature studies.

3.2.3.1 Passive network

This matrix is the traditional nodal admittance matrix used in power flow solution.

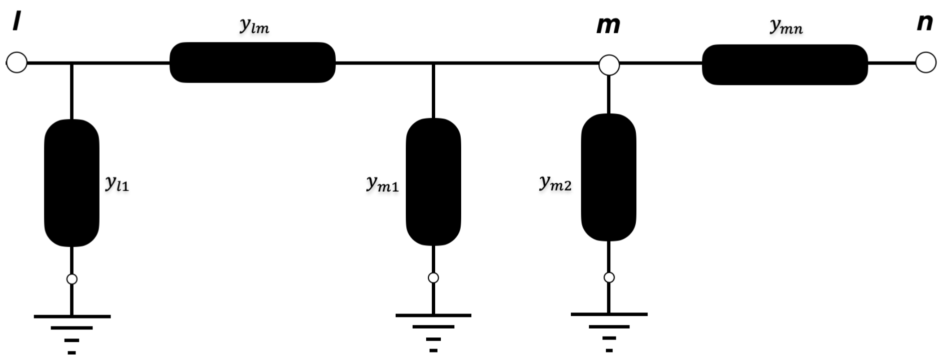


Figure 3.2: Primitive admittance matrix circuit example

$$Y_{passive} = \begin{bmatrix} Y_{11} & \dots & Y_{1m} & \dots & Y_{1N} \\ \vdots & \ddots & \vdots & & \vdots \\ Y_{m1} & \dots & Y_{mm} & \cdots & Y_{mN} \\ \vdots & & \vdots & \ddots & \vdots \\ Y_{N1} & \dots & Y_{Nm} & \dots & Y_{NN} \end{bmatrix} \tag{3.13}$$

The matrix are symmetrical ($Y_{mN} = Y_{Nm}$), and often sparse, so matrix techniques that make use of these properties provides computational gain.

The matrix is formed as follows

$$Y_{ii} = \sum_{k=1,\, k\neq i}^{n} y_{ik} + y_i^{\mathrm{sh}}, \tag{3.14}$$

$$Y_{ik} = -\, y_{ik}, \quad i \neq k \tag{3.15}$$

where y_{ik} is the admittance of the branch connecting buses i and k and y_i^{sh} represents the shunt admittance connected to bus i.

3.2.3.2 Primitive admittance matrix

In the following section, the electrical model of the equipments will be presented, but before, a convenient way to construct an admittance matrix will be presented next. Often denominated as primitive admittance matrix [57], it consists of constructing a 2×2 or 1×1 admittance matrix for each element and then lumping the admittances on the respective bus.

Consider the example in Figure 3.2. It consists of a transmission line between buses l and m, a shunt element in bus m and a transformer between buses m and n.

It is important to maintain the node reference for the primitives. The primitive matrices of the three elements are

$$Y_{line}^{prim} = \begin{array}{cc} \begin{array}{cc} l & m \end{array} & \\ \begin{pmatrix} y_{lm} + y_{l1} & -y_{lm} \\ -y_{lm} & y_{lm} + y_{m1} \end{pmatrix} & \begin{array}{c} l \\ m \end{array} \end{array}$$

$$Y_{shunt}^{prim} = \begin{array}{cc} m & \\ (y_{m2}) & m \end{array}$$

$$
Y^{prim}_{transformer} = \begin{array}{cc} \begin{array}{cc} m & n \end{array} & \\ \begin{pmatrix} y_{mn} & -y_{mn} \\ -y_{mn} & y_{mn} \end{pmatrix} & \begin{array}{c} m \\ n \end{array} \end{array}
$$

The final matrix is simply the summation of the admittances in the right index position.

$$
Y^{prim} = \begin{array}{cc} \begin{array}{ccc} l & m & n \end{array} & \\ \begin{pmatrix} y_{lm} + y_{l1} & -y_{lm} & 0 \\ -y_{lm} & y_{lm} + y_{m1} + y_{m2} + y_{mn} & -y_{mn} \\ 0 & -y_{mn} & y_{mn} \end{pmatrix} & \begin{array}{c} l \\ m \\ n \end{array} \end{array} \tag{3.16}
$$

3.2.3.3 Faulty network

The short circuit admittance matrix (Y_{SCC}) is an extension of the passive admittance matrix ($Y_{passive}$).

Considering the matrix shown in Equation 3.4 and the primitive admittance matrix concept, most of the short circuit admittance matrix construction is already covered. One might notice that, starting from a $Y_{passive}$ matrix and primitive admittance matrices of the active elements, as highlighted in Figure 3.3, the construction of the final Y_{SCC} is simply the lumping of the admittances in the correct node index, similar as performed in the Figure 3.2 example, Equation 3.16.

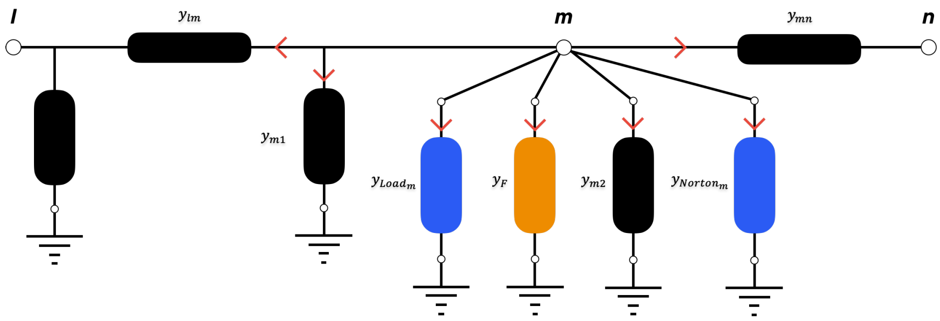


Figure 3.3: A section of an example network, showing the passive elements in black, active elements in blue and fault impedance in orange

Referring Figure 3.3

$$
\begin{aligned}
Y_{mm} &= (y_{lm} + y_{m1} + y_{m2} + y_{mn}) + (y_{Load_m} + y_{Norton_m}) + y_F \\
&= (y_{passive,mm}) + (y_{active,mm}) + y_F
\end{aligned}
$$

Where y_{Norton_m} is a generic impedance from the Norton equivalent model, thus, it could be a load, a synchronous machine, a converter, etc. Note that the off-diagonal elements Y_{lm}, Y_{mn} remain unchanged since the active elements are shunt connected.

As the active elements primitives are 1×1 matrices (as they are shunt connected), making changes in the short circuit matrix (Y_{SCC}) is cheap, since the heavy cost is the construction the $Y_{passive}$ matrix. Switching active elements on/off are also achieved in a simple manner, as they can be accessed individually.

We have seen how the admittance matrix is built through the addition of individual elements, the next section presents how each element is computed for the Short-Circuit Calculation proposed in this work.

3.3 Electrical Models

We have seen in Equation 3.3 that to formulate the short circuit problem it is needed two pieces of information: the short circuit admittance matrix (Y_{SCC}) and the nodal current injection ($I_{injection}$).

In this section, each electrical element considered in this work method will be covered. For each element, their primitive matrix and current injection model are to be presented. Enough information to construct $Y_{passive}$, Y_{SCC}, perform power flow solutions and ultimately, perform short circuit calculation.

3.3.1 Passive elements

Passive elements model are represented only by their primitive admittance matrix (Y^{prim}) and have no current injection.

3.3.1.1 Transmission Lines

Implemented as the π circuit, Figure 3.4.

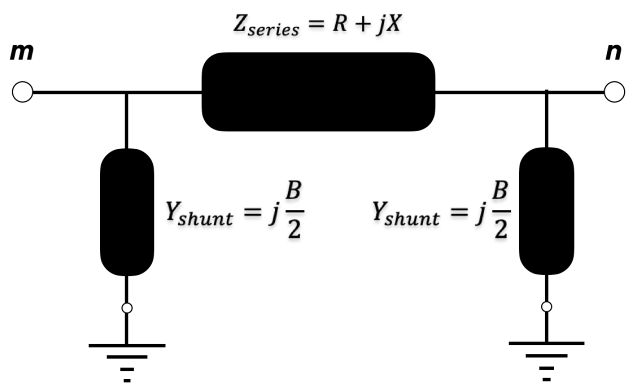


Figure 3.4: Transmission line model

$$Y_{line} = \begin{bmatrix} Y_{mm} & Y_{mn} \\ Y_{nm} & Y_{nn} \end{bmatrix} = \begin{bmatrix} \frac{1}{Z_{series}} + j\frac{B}{2} & -\frac{1}{Z_{series}} \\ \frac{1}{Z_{series}} & \frac{1}{Z_{series}} + j\frac{B}{2} \end{bmatrix}$$

Note that this is a positive sequence implementation and implies balanced, uncoupled lines, as already discussed previously.

3.3.1.2 Transformers

Transformers are also modeled in the positive sequence, as shown in Figure 3.5. As stated previously, the phase shifts are neglected. Additionally, the magnetizing impedance are also neglected, often common in short circuit calculations [58] due to the relative magnitude of the magnetizing current in comparison to the winding current.

$$Y_{transformer} = \begin{bmatrix} Y_{mm} & Y_{mn} \\ Y_{nm} & Y_{nn} \end{bmatrix} = \begin{bmatrix} \frac{1}{Z_{series}} & -\frac{1}{Z_{series}} \\ \frac{1}{Z_{series}} & \frac{1}{Z_{series}} \end{bmatrix}$$

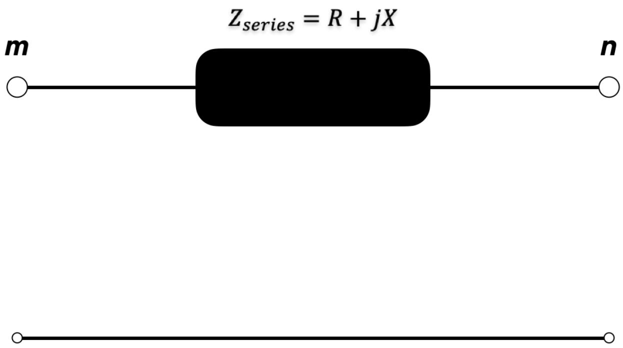


Figure 3.5: Transformer model

3.3.1.3 Uncontrolled reactive shunt elements

These are capacitor and reactor banks directly connected to the network, as shown in Figure 3.6. In contrast to controlled reactive devices such as Static Var Compensator (SVC) and STATCOM, the elements characterized in this work as uncontrolled reactive shunt elements does not have a power electronics interface with the grid, thus, they are represented as linear (passive) elements.

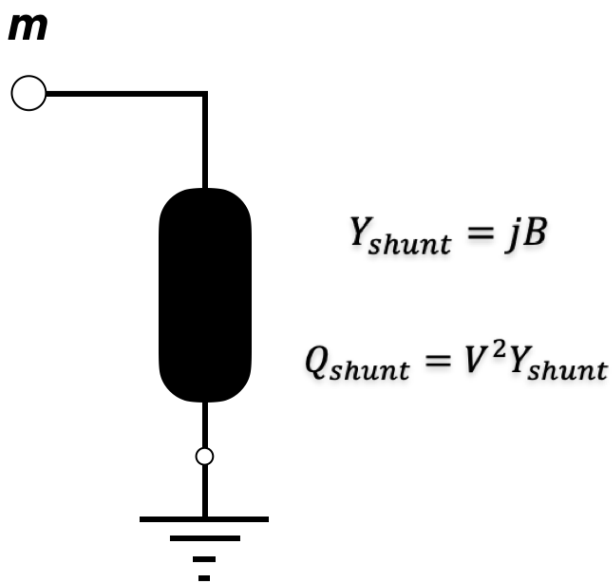


Figure 3.6: Uncontrolled reactive shunt model

$$Y_{shunt\,element} = \left[Y_{mm}\right] = \left[jB\right]$$

3.3.2 Active elements

Active elements model, differently from passive elements, have current injections in addition to their primitive admittance matrix (Y^{prim}) for a short circuit calculation.

The current injection can be a constant value (synchronous/asynchronous machines), nonlinear voltage dependent (full converters). The following sections cover Type A, Type B and Type D production facilities from Section 2.2, although Doubly-fed Induction Generator (Type C) are not included in this work, the proposed method supports its future inclusion.

Loads might be modeled in different ways [59], constant impedance, constant current injection to constant power, ZIP, complex model, etc. Although some method neglect loads, others concludes that neglecting may lead to inaccurate results [48], the challenge however, is to accurately represent the load.

3.3.2.1 Loads

The model of choice is the constant impedance, and its equivalent circuit is shown in Figure 3.7. The load current will decrease as the terminal voltage decreases. The impedance is fixed and the current and voltage relation are completely described by the impedance, in this case then, the load behaves as a passive element without active current injection.

Worth noticing that this work method is agnostic regarding the load model, therefore, a variety of models (linear and nonlinear) could be implemented.

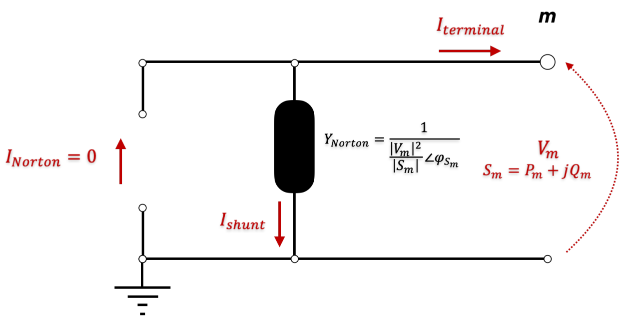


Figure 3.7: Constant impedance load model

The load impedance is calculated with

$$Z_{load} = \frac{|V_m^2|}{|S_m|}\angle angle(S_m) \tag{3.17}$$

Where V_m and S_m comes from power flow solution or a supervisory system measurement.

The load primitive admittance matrix to be included at the short circuit admittance matrix Y_{SCC} is defined as

$$Y_{load} = \begin{bmatrix} Y_{mm} \end{bmatrix} = \begin{bmatrix} \frac{1}{Z_{load}} \end{bmatrix}$$

The load fault model response to different terminal voltages are shown in Figure 3.8. As for this example, the load is a constant impedance model, therefore when a solid fault happens in the load terminal ($V_m = 0$), the element makes no contribution, while for voltages around the nominal ($V_m \approx 1$), the current on the element is its pre-fault current.

3.3.2.2 Synchronous Generators

Conventional protection design consider the synchronous generator in the subtransient period, thus represented by its subsynchronous reactance on the Thévenin equivalent model as in Figure 3.9. As discussed previously, asynchronous (induction) generators directly connected to the grid are often modeled similarly to the synchronous machine [21, 37, 38]. Thus, this section contains the model for production facilities **Type A** and **Type B** from Section 2.2.

The machine internal voltage E_g is further assumed constant during the short-circuit and can be calculated by

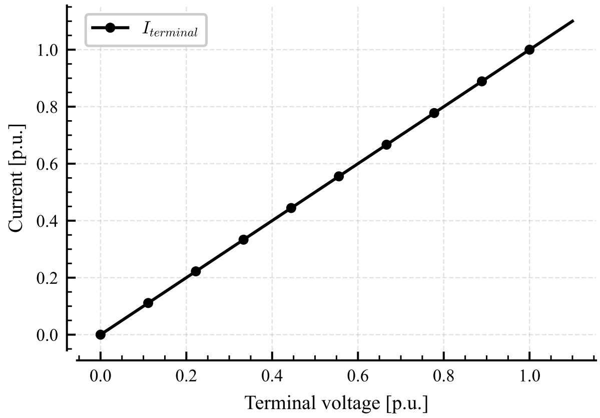


Figure 3.8: Load fault model response

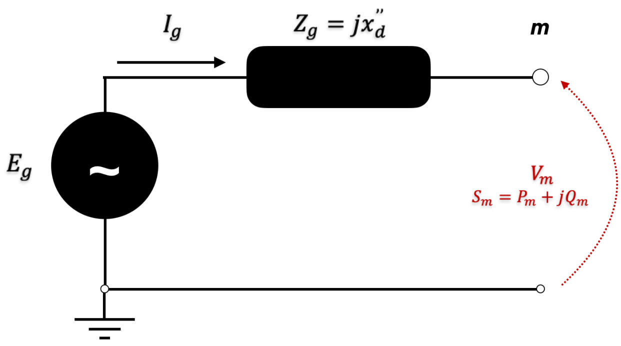


Figure 3.9: Generator Thévenin model

$$E_g = V_m^{pf} + Z_g I_g^{pf}$$

the notation pf indicates the pre-fault state.

$$I_g^{pf} = \left(\frac{S_m^{pf}}{V_m^{pf}}\right)^* \tag{3.18}$$

Where S_m and V_m are the terminal power and voltage of the machine, obtained by a power flow solution or acquired on a supervisory system. The machine internal voltage is then

$$E_g = V_m^{pf} + jx_d'' \left(\frac{S_m^{pf}}{V_m^{pf}}\right)^* \tag{3.19}$$

resulting in

$$I_{Norton} = \frac{E_g}{jx_d''} \tag{3.20}$$

$$Y_{Norton} = \frac{1}{jx_d''} \tag{3.21}$$

The machine Norton equivalent, shown in Figure 3.10, will be the one used on the short circuit calculation.

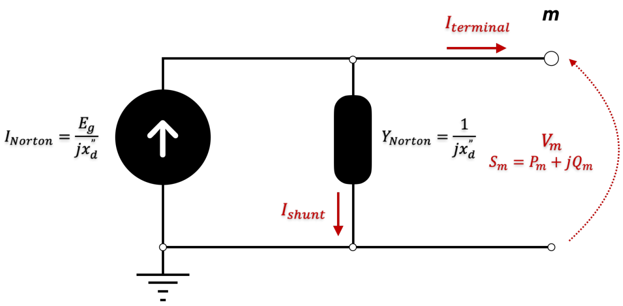


Figure 3.10: Generator Norton model

The synchronous machine primitive admittance matrix to be included at the short circuit admittance matrix Y_{SCC} is defined as

$$Y_{SG} = [Y_{mm}] = [Y_{Norton}]$$

The synchronous generator model response to different terminal voltages are shown in Figure 3.11. When a solid fault happens at its terminals ($V_m = 0$), no current flows through the shunt branch, thus the machine has its largest contribution to the short circuit. At the nominal voltage ($V_m \approx 1$), the terminal current converges to its pre-fault current.

3.3.2.3 Full converters

For starters, it is important to briefly discuss the timescale of the converters' dynamics for short circuit modeling and the coherence with the synchronous generator model. Synchronous generators substransient state may last up to $40 \sim 50\,ms$ while the converters fault steady state begins at $40 \sim 50\,ms$ [9]. For this work, it is considered this intersection,

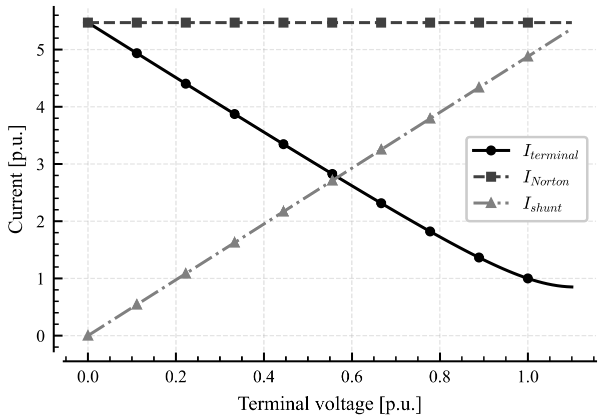


Figure 3.11: Synchronous generator fault model response

therefore, the synchronous machine is modeled in the substransient state, as shown previously, while the converter is modeled in the steady state. This section contains the model for production facilities **Type D** from Section 2.2.

The converter equivalent circuit is shown in Figure 3.12. Similarly as in [23], it is assumed that full converters positive sequence shunt impedance as infinite, i.e. admittance is zero. Therefore, full converters do not change the short circuit admittance matrix Y_{SCC}.

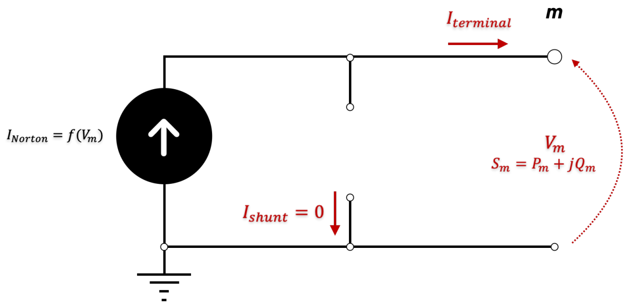


Figure 3.12: Full converter model

The converter model is for a grid following converter (GFL), and its controlled current source is modeled in the steady state fault stage [9], using reactive current prioritization with additional reactive current injection for voltage support (K-factor) [13]. The maximum converter current (I_{max}) is set to 1.2 times its nominal current. During a short circuit, the current is dependent of the actual bus voltage (V_m), the pre-fault voltage (V_m^{pf}) and pre-fault apparent power (S_m^{pf}). The previous parameters are measurements in the point of common coupling (PCC) of the production facility. The governing equations are presented below.

First, the pre-fault condition is calculated

$$I_m^{pf} = \left(\frac{S_m^{pf}}{V_m^{pf}}\right)^* \tag{3.22}$$

Next, the angle of the actual bus voltage is taken

$$\theta = \arg V_m \tag{3.23}$$

Apply rotation to align current phasor with terminal voltage. This step is similar to applying Park transform and, given the balanced system assumption, can be simplified as follows:

$$I_{dq} = I_m^{pf} \cdot e^{-j\theta} = I_d + jI_q \tag{3.24}$$

As I_{dq} is aligned with the terminal voltage, its real part $\Re(I_{dq}) = I_d$ is proportional to active power injection and the imaginary part $\Im(I_{dq}) = I_q$ to reactive power injection at the point of common coupling.

The next step applies the additional reactive current injection (ΔI_q) for voltage support. As the K-factor (K) is normalized by the device nominal current (I_n) [13], an additional step is needed, to refer ΔI_q to the system common reference (I_b).

$$\Delta I_q = K(|V_m| - |V_m^{pf}|)\left(\frac{I_n}{I_b}\right) \tag{3.25}$$

K usually varies from 0 to 10 with a common value of 3 [13]. The nonlinearity at the short-circuit calculation comes from Equation 3.25, as it introduces the dependence of the converter current ($I_{converter}$) to the terminal voltage (V_m).

Define the requested reactive current as

$$I_q^{requested} = I_q + \Delta I_q \tag{3.26}$$

As I_{max} is a multiple of the nominal current (I_n), it needs to be refereed to the system common base

$$I'_{max} = I_{max}\left(\frac{I_n}{I_b}\right) \tag{3.27}$$

And then apply reactive current prioritization (q-prioritization). It is important to retain the sign of the current since it determines the absorption/injection characteristics.

$$I_q^* = \begin{cases} -I'_{max}, & \text{if } I_q^{requested} < -I'_{max} \\ I'_{max}, & \text{if } I_q^{requested} > I'_{max} \\ I_q^{requested}, & \text{otherwise} \end{cases} \tag{3.28}$$

apply current limitation (saturation) and compute the real component of the converter current

$$I_d^{max} = \sqrt{(I'_{max})^2 - (I_q^*)^2} \tag{3.29}$$

$$I_d^* = \begin{cases} -I_d^{max}, & \text{if } I_d < -I_d^{max} \\ I_d^{max}, & \text{if } I_d > I_d^{max} \\ I_d, & \text{otherwise} \end{cases} \tag{3.30}$$

Finally, compute the complex converter voltage and reestablish the phasor rotation made in Equation 3.24

$$I_{converter} = (I_d^* + jI_q^*) \cdot e^{j\theta} \tag{3.31}$$

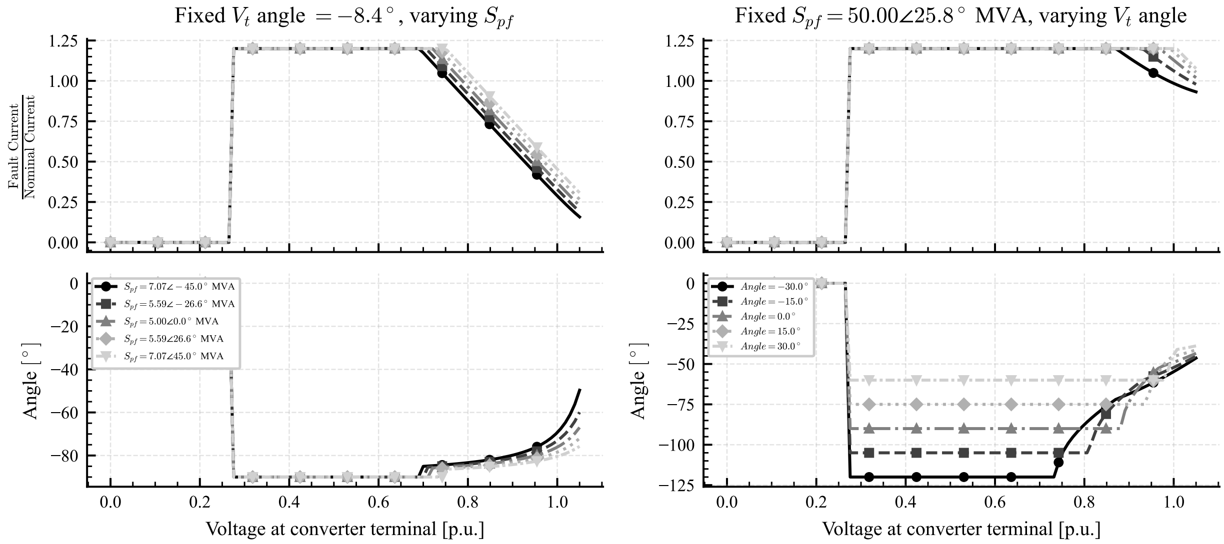
$I_{converter}$ is the Norton current source I_{Norton} from Figure 3.12.

The response of the converter under fault, for different pre-fault (denoted with *pf*) operating points, are shown in Figure 3.13(a). The voltage at converter terminal, the x-axis, varies with a fixed angle from $0.0\angle 30°$ to $1.05\angle 30°$.

Note that the converter stays online until its terminal voltages reaches a minimum value, as established in the low-voltage ride through requirement from [19].

Changing the angle of the voltage at the converter terminals will also change the converter response, as shown in Figure 3.13(b).

Lastly, the converter response to different K-factors are shown in Figure 3.14.



(a) (b)

Figure 3.13: Converter fault model response. Pre-fault voltage $V_m^{pf} = 1.12\angle -8.42^\circ$ p.u. and K-factor $K = 3.0$.

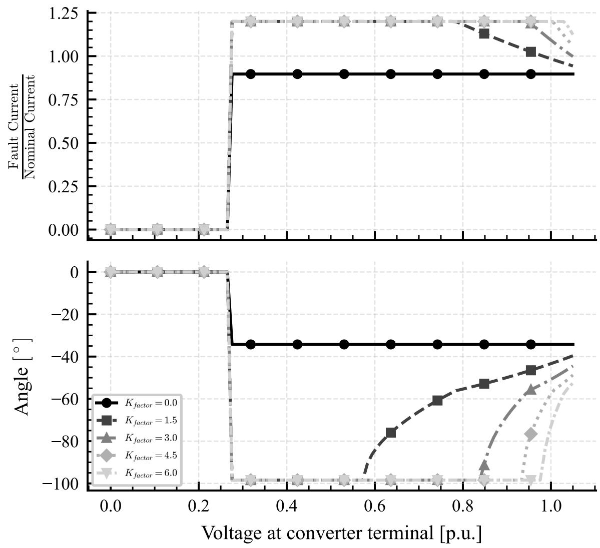


Figure 3.14: Converter fault model response. Pre-fault voltage $V_m^{pf} = 1.12\angle -8.42^\circ$ p.u. and Pre-fault power $S_m^{pf} = 50.00\angle 25.84^\circ$ p.u.

3.3.2.4 External Grids

The external grid fault model is a network equivalent based on static network reduction, where the objective is to reproduce the electrical behavior of the original network at the boundary interface with sufficient accuracy [20]. Ultimately in the short circuit calculation scope, this is achieved by the equivalent network short circuit capacity at the boundary node [23].

Let the variable of interest be the external grid equivalent impedance ($Z_t h$), depending on the available information, it can be achieved by three different ways.

First case: The equivalent resistance and reactance of the external grid are known, and the equivalent impedance is given by:

$$Z_{th} = R_{th} + jX_{th}$$

Second case: The short-circuit capacity (S_{SC}) is known, and the equivalent impedance is given by:

$$|Z| = \frac{V_{nominal}^2}{|S_{SC}|}$$

$$X_{th} = 0.995|Z| \tag{3.32}$$

$$R_{th} = 0.1X_{th} \tag{3.33}$$

$$Z_{th} = R_{th} + jX_{th}$$

The constants 0.995 and 0.1 from the previous Equation 3.32 and Equation 3.33 are in accordance to the IEC 60909 [23] in cases where the X/R ratio are unknown. If X/R of the equivalent is known, it should be applied instead.

Third case: The symmetrical short-circuit current (I''_{k3}) is known, and the equivalent impedance is given by:

$$|Z| = \frac{V_{nominal}}{\sqrt{3}I''_{k3}}$$

Then the same steps as in Equation 3.32 and Equation 3.33 are applied.

Note that the equivalent impedance is constant for every case. Then, to integrate the external grid model into the short circuit calculation method, their Norton equivalent (Figure 3.15) is necessary.

$$Y_{Norton} = \frac{1}{Z_{th}} \tag{3.34}$$

$$E_{th} = V_m^{pf} + Z_{th}\left(\frac{S_m^{pf}}{V_m^{pf}}\right)^* \tag{3.35}$$

Then

$$I_{Norton} = \frac{E_{th}}{Z_{th}} \tag{3.36}$$

It can be observed that the Norton equivalent of the external grid fault model resembles that of a synchronous machine, in that the current source is assumed to deliver a constant current during a short circuit.

Referring to Figure 3.15, the short circuit current contribution of an external grid ($I_{terminal}$) is limited by the current through the Norton admittance (I_{shunt}) since

$$I_{terminal} = I_{Norton} - I_{shunt}$$

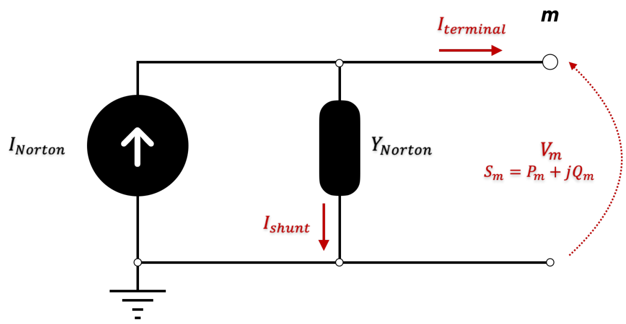


Figure 3.15: External grid model

The external grid fault model response to different terminal voltages are shown in Figure 3.16. When a solid fault happens in the boundary node ($V_m = 0$) no current flows through the shunt branch, thus the external grid element has its largest contribution to the short circuit. At the nominal voltage ($V_m \approx 1$), the terminal current converges to its pre-fault current.

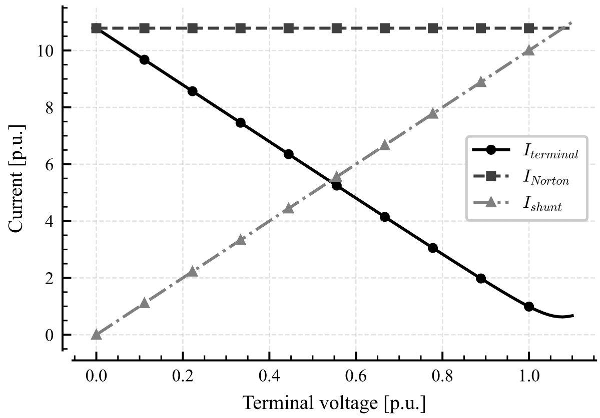


Figure 3.16: External grid fault model response

3.4 Workflow for testing, validation and deployment

Figure 3.17 shows a practical workflow for implementing the presented method. The short circuit calculation ("Calculate I_{SC}") in covered in Section 3.2.1, The pre-fault state ("Solve power flow") in Section 3.2.2. The steps involving matrix manipulation - "Compose $Y_{passive}$", "Add Y^{prim}" and "Add fault bus $'m'$" are covered in Section 3.2.3.

Data reading, "Read data" in Figure 3.17, in this work is done by parsing and processing a JSON file using Object-oriented programming for convenience, although other approaches impose no limitation to the method. The data required are commonly found data in power system studies and operational measurement, Table A.1 in the Appendix refers to Svenska Kraftnät Power System Modeling Handbook [60] and refers to the Common Information Model (CIM), a group of standards for exchanging network data.

Operation conditions are user-defined and provides the interface to emulate different conditions in an electric power system. Changes in generation, load and contingencies propagates on the model through this process. Note the processing sequence and how changes on operation conditions influences the pre-fault state and consequently, the short circuit calculation. This propagation chain is what make the method suitable for continuous short circuit calculation, as it provides a comprehensive linkage between the power grid state and the short circuit estimation.

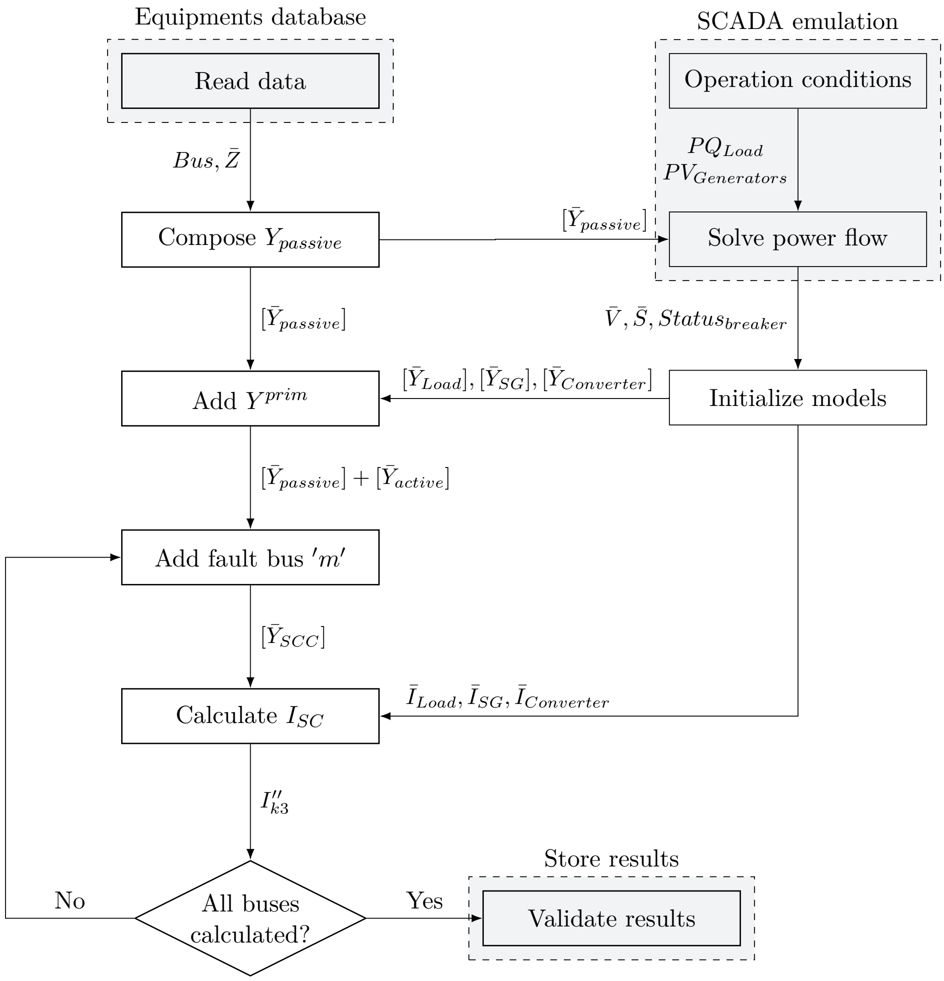


Figure 3.17: Flowchart for the short circuit calculation method

The validation of the present method is conducted using third-party commercial software such as PowerWorld [61], pandapower [62] or similar on the IEEE 14-bus test case and modification of it.

Chapter 4

Results and Discussion

This chapter aims to show the ability of the proposed methodology to provide accurate results. Validations against third-party software, and simulations considering different scenarios and assumptions are included in this chapter.

4.1 Reference Network and Method Validation

In this section, the short circuit current and power will be computed using commercial software and then compared to the proposed method from this work. The simulations use public available information for reproducibility.

4.1.1 Definition of the reference network

The reference network utilized in this result section is the IEEE 14-bus test case [63]. The single line diagram of the network is shown in Figure 4.1.

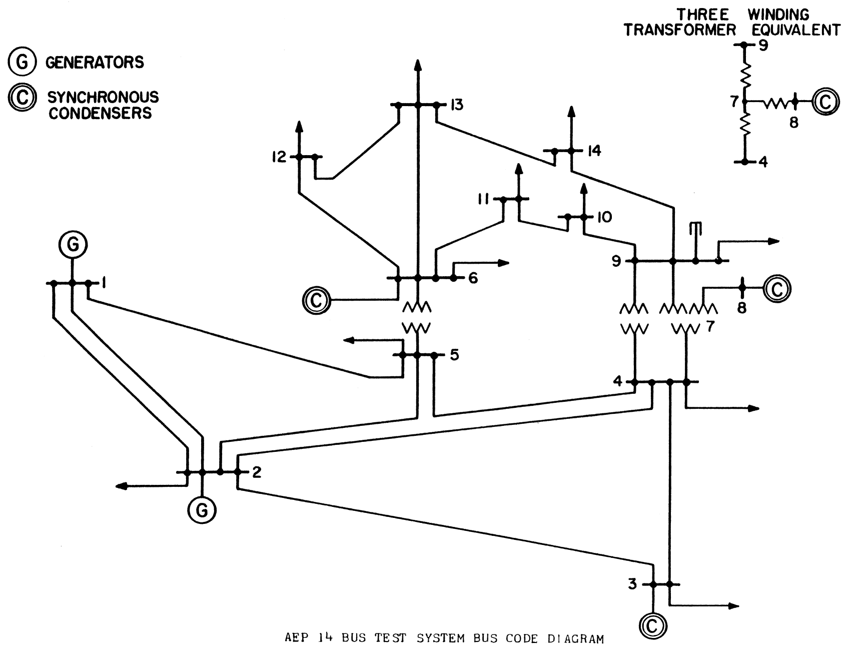


Figure 4.1: IEEE 14-bus test case. Reproduction from [63].

The network is balanced and contains 14 buses, 1 shunt capacitor, 5 generators (2 generators and 3 synchronous condensers) and 11 loads. In the default operating point, the system loading are $P_{generation} = 272.4$ MW, $P_{load} = 259.0$ MW and $Q_{load} = 73.5$ Mvar. Table 4.1 and Table 4.2 presents the generators and loads for the reference network, respectively.

Table 4.1: Generators of the reference network.

Generator	Bus	P (MW)	V (p.u.)
G1	1	232.40	1.060
G2	2	40.00	1.045
G3	3	0.00	1.010
G6	6	0.00	1.070
G8	8	0.00	1.090

Generators G3, G6 and G8 from Table 4.1 are synchronous condensers, thus their active power is zero.

Table 4.2: Loads of the reference network.

Load	Bus	P (MW)	Q (MVAr)
Load 2	2	21.70	12.70
Load 3	3	94.20	19.00
Load 4	4	47.80	-3.90
Load 5	5	7.60	1.60
Load 6	6	11.20	7.50
Load 9	9	29.50	16.60
Load 10	10	9.00	5.80
Load 11	11	3.50	1.80
Load 12	12	6.10	1.60
Load 13	13	13.50	5.80
Load 14	14	14.90	5.00

4.1.2 Validation of the continuous short circuit estimation method

This section presents the short circuit values obtained from the proposed method, and then compared to the values from PowerWorld simulation. Three distinct scenarios are evaluated, first at the default operating point, second with the three synchronous condensers disconnected and third with the generator from bus two disconnected.

4.1.2.1 Default operating point - reference case

In this scenario, all generators are connected, and the loads are held at $S_{load} = 259.0 + j73.5$ MVA. The power flow solution from the proposed method is first compared to the original reference [63] in Table 4.3. The largest difference in voltage or angle is less than 0.2%.

Figure 4.2 shows the short circuit current in per-unit in each bus of the network. It can be seen that buses electrically closer (lower impedance path) to bus 1, which contains the network's largest generator, have higher short circuit level. The asymmetrical short circuit level profile throughout the network is a consequence of its specific topology and the

Table 4.3: Power flow results from the reference network and the proposed method.

	Voltage Magnitude (p.u.)			Voltage Angle (°)		
Bus	PowerWorld	Proposed	Δ (%)	PowerWorld	Proposed	Δ (%)
1	1.06	1.06	0.00	0.00	0.00	0.00
2	1.05	1.05	0.00	-4.98	-4.98	+0.05
3	1.01	1.01	0.00	-12.72	-12.73	+0.04
4	1.02	1.02	-0.13	-10.33	-10.31	-0.17
5	1.02	1.02	-0.05	-8.78	-8.77	-0.07
6	1.07	1.07	0.00	-14.22	-14.22	+0.01
7	1.09	1.09	0.00	-13.36	-13.36	-0.08
8	1.06	1.06	-0.05	-14.94	-14.94	-0.01
9	1.05	1.05	0.00	-15.10	-15.10	-0.02
10	1.06	1.06	-0.01	-14.79	-14.79	-0.02
11	1.06	1.06	+0.02	-15.07	-15.08	+0.04
12	1.05	1.05	+0.04	-15.16	-15.16	-0.02
13	1.06	1.06	-0.01	-14.94	-14.94	-0.01
14	1.04	1.04	-0.05	-16.04	-16.03	-0.05

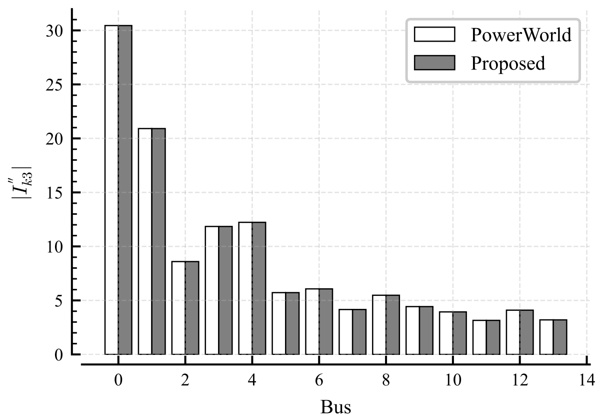


Figure 4.2: Per-bus short-circuit current, base case.

characteristics of its elements. Thus, short circuit estimation methodologies that do not encode these parameters could not estimate short circuit with sufficient precision.

Furthermore, Table 4.4 shows that, for this default scenario, the proposed method achieves the same solution as PowerWorld.

Table 4.4: Short-circuit current validation against PowerWorld, base case.

Bus	PowerWorld (p.u.)	Proposed (p.u.)	Δ (%)
1	30.45	30.45	-0.00
2	20.91	20.91	-0.00
3	8.60	8.60	-0.00
4	11.85	11.85	-0.00
5	12.23	12.23	-0.00
6	5.72	5.72	-0.00
7	6.07	6.07	-0.00
8	4.16	4.16	-0.00
9	5.48	5.48	-0.00
10	4.43	4.43	+0.00
11	3.94	3.94	+0.00
12	3.15	3.15	+0.00
13	4.10	4.10	+0.00
14	3.20	3.20	-0.00

4.1.2.2 Synchronous condensers disconnected

In this scenario, all synchronous condensers are disconnected (only generators at bus 1 and 2 connected) and the loads are held at the same $S_{load} = 259.0 + j73.5$ MVA.

The error found in this condition is presented in Table 4.5. It was observed a maximum error of 2.32% in bus 9. This can be result of controls not implemented in the present method which are presented in the PowerWorld, specially since in this case, the generator in bus 2 has reached its reactive limit and as the synchronous condensers are now disconnected, the bus voltages across the network are significant smaller.

Table 4.5: Short-circuit current validation against PowerWorld, only G1 and G2 active.

Bus	PowerWorld (p.u.)	Proposed (p.u.)	Δ (%)
1	28.88	29.00	+0.40
2	18.29	18.47	+0.96
3	6.20	6.31	+1.65
4	9.19	9.36	+1.80
5	9.79	9.95	+1.61
6	3.81	3.89	+1.98
7	4.21	4.29	+1.89
8	2.41	2.44	+1.17
9	4.02	4.12	+2.32
10	3.36	3.43	+2.03
11	2.98	3.02	+1.65
12	2.41	2.44	+1.39
13	3.00	3.06	+1.90
14	2.53	2.57	+1.87

4.1.2.3 Generator 2 disconnected

In this scenario, only generator of bus 2 is disconnected (generators of bus 1, 3, 6 and 8 connected) and the loads are held at the same $S_{load} = 259.0 + j73.5$ MVA.

Table 4.6 shows the error found for this condition.

Table 4.6: Short-circuit current validation against PowerWorld, G2 disconnected.

Bus	PowerWorld (p.u.)	Proposed (p.u.)	Δ (%)
1	26.31	26.30	-0.03
2	13.60	13.58	-0.09
3	7.76	7.73	-0.39
4	10.12	10.11	-0.12
5	10.47	10.46	-0.09
6	5.47	5.47	-0.04
7	5.74	5.73	-0.07
8	4.06	4.06	-0.03
9	5.20	5.20	-0.06
10	4.25	4.25	-0.05
11	3.80	3.80	-0.03
12	3.07	3.07	-0.02
13	3.96	3.96	-0.03
14	3.10	3.10	-0.03

This section presented the results obtained from the proposed method when compared against a commercial software, however, in the sections to follow, the results obtained from the method will be compared against itself, given different scenarios.

4.2 Effect of System Operating Conditions on Short-Circuit Capacity

In this section, simulations are conducted on the reference network while changing loading, and topology. The results are then compared against the default operation point (called **Reference** in this section, as presented in as in Section 4.1.2.1) to evaluate the marginal effect of these conditions on the short circuit level of the buses of the network.

4.2.1 Low load

The network loading is evenly reduced through applying a scaling factor of 0.3 to all loads. This scenario is called **Simulated** in this present section.

Table 4.7 shows the summary of the scenarios under comparison. In the simulated scenario, the loads complex power is reduced by 70% with constant power factor.

In Figure 4.3 and complementary in Table 4.8 its observed that even for a large decrement in the network loading and thus in the generation, the order of impact in the short circuit level across the relatively small.

The impact can be evaluated through Equation 3.19 and Equation 3.20, as the current injection from the synchronous machine model (I_{Norton}) have two components, one that is inversely proportional to the impedance and another proportional to the complex pre-fault power. For an example machine with $X''_d = 0.2$ and $V_m^{pf} = 1.0$ the difference between

Table 4.7: Pre-fault condition for default operation point and low load.

Metric	Reference	Simulated	Δ (%)
Total generation (MW)	272.39	78.63	-71.13
Total renewable (MW)	0.00	0.00	
Renewable share (%)	0.00	0.00	
Total load (MW)	259.00	77.70	-70.00
Losses (%)	4.92	1.19	-75.84

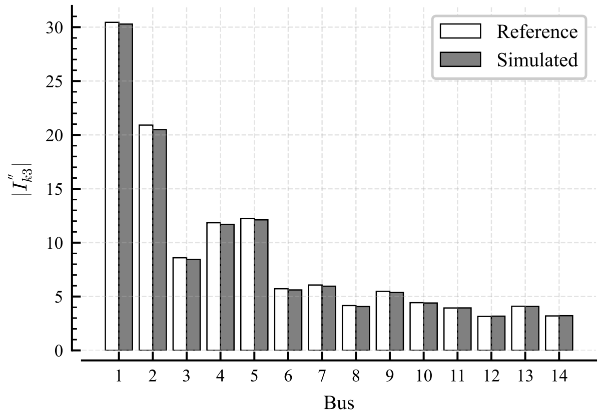


Figure 4.3: Per-bus short-circuit current, low load condition.

the full-load condition ($|S_m|^{pf} = 1.0$ p.u., $I_{Norton} = 6.0$ p.u.) and the no-load condition ($|S_m|^{pf} = 0.0$ p.u., $I_{Norton} = 5.0$ p.u.) is a reduction of 17%.

The different loading scenarios highlights that most of the short-circuit contribution from a synchronous machine comes passively from its impedance (e.g. their inherent property [1]) and also that larger machines (lower X''_d) are less sensible to this loading behavior.

Additionally, when analyzing the network topology, specifically regarding the location and size of the loads as shown in Table 4.2, no clear correlation between it and the results from Table 4.8 are to be seen, as some buses experience an increase in the short-circuit current while other a reduction. Therefore, in addition to the loading of generators, the pre-fault condition of the network and the load short circuit model itself (Section 3.3.2.1) are further dependencies of the short circuit value.

However, as it is observed in the following sections, for this specific network, the network loading are not the main factor that drives large changes or discrepancies in the short circuit calculation.

4.2.2 High load

The network loading is now evenly increased through applying a scaling factor of 1.7 to all loads. This scenario is called **Simulated** in this present section.

Table 4.9 shows the summary of the scenarios under comparison. In the simulated scenario, the loads complex power is increased by 70% with constant power factor. Note that, similarly that as from Table 4.7, the percent losses do not change linearly since $P_{loss} \propto I^2$.

Different from the low load scenario in Figure 4.3, the high load case in Figure 4.4 already shows a notable difference in the short circuit levels. During high loading, the generator may reach its under excitation limiter (UEL) or over excitation limiter (OEL) and cannot further

Table 4.8: Numerical values of per-bus short-circuit current, low load condition.

Bus	Reference (p.u.)	Simulated (p.u.)	Δ (%)
1	30.45	30.29	-0.52
2	20.91	20.50	-1.98
3	8.60	8.44	-1.84
4	11.85	11.70	-1.24
5	12.23	12.12	-0.93
6	5.72	5.61	-1.91
7	6.07	5.96	-1.83
8	4.16	4.07	-2.16
9	5.48	5.37	-1.92
10	4.43	4.40	-0.66
11	3.94	3.94	+0.14
12	3.15	3.18	+0.78
13	4.10	4.08	-0.43
14	3.20	3.22	+0.65

Table 4.9: Pre-fault condition for default operation point and high load.

Metric	Reference	Simulated	Δ (%)
Total generation (MW)	272.39	501.16	+83.99
Total renewable (MW)	0.00	0.00	
Renewable share (%)	0.00	0.00	
Total load (MW)	259.00	440.30	+70.00
Losses (%)	4.92	12.14	+147.00

support reactive power to the grid, thus the voltage on the bus decreases. As discussed in the previous section, in addition to the pre-fault power, the pre-fault voltage is a parameter of the synchronous machine short circuit current injection, and in this case a more critical one, as seen in Equation 3.19 and Equation 3.20.

The short circuit current on bus 1 increases because this bus is the slack bus and the generator connected to it is the largest in the network. These means that the bus does not suffer voltage deviation and the additional power that the local generators cannot supply are requested from the generator in that bus. As one might conclude from the results so far that an increase in the generator loading for a fixed terminal voltage will increase the short circuit current. As remarked, the previous conclusion is easily observed at the terminals of the machine, however the propagation of the short circuit injection throughout the network is determined by its topology.

Then in Table 4.10, a trend is observed with the buses that are further from the slack bus experiencing a higher decrease in the short circuit capacity. Local generation would increase these buses short circuit capacity, not solely through the increment of the voltage, but also by the additional short circuit capacity that these additional generations would include, i.e. the synchronous machine inherent property, as mentioned in [1] and further discussed in this work.

4.2.3 Large load disconnection

So far, the loading change was uniform across the network, thus, in this scenario the largest load (load in bus 3) will be disconnected, and the other remained in their default operation

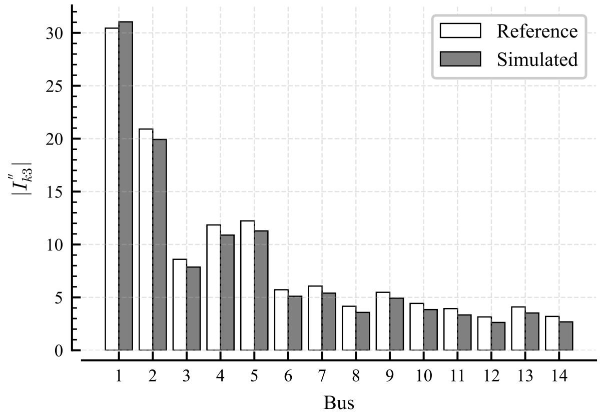


Figure 4.4: Per-bus short-circuit current, high load condition.

Table 4.10: Numerical values of per-bus short-circuit current, high load condition.

Bus	Reference (p.u.)	Simulated (p.u.)	Δ (%)
1	30.45	31.04	+1.96
2	20.91	19.92	-4.72
3	8.60	7.86	-8.51
4	11.85	10.89	-8.09
5	12.23	11.28	-7.75
6	5.72	5.11	-10.77
7	6.07	5.40	-10.99
8	4.16	3.58	-13.98
9	5.48	4.92	-10.12
10	4.43	3.85	-13.17
11	3.94	3.34	-15.16
12	3.15	2.63	-16.61
13	4.10	3.53	-13.96
14	3.20	2.69	-16.03

point. This scenario is called **Simulated** in this present section.

As seen in Table 4.11 the load in bus 3 represents more than a third of the network power.

Table 4.11: Pre-fault condition for default operation point and large load disconnection.

Metric	Reference	Simulated	Δ (%)
Total generation (MW)	272.39	169.90	-37.63
Total renewable (MW)	0.00	0.00	
Renewable share (%)	0.00	0.00	
Total load (MW)	259.00	164.80	-36.37
Losses (%)	4.92	3.00	-38.97

Then at Table 4.12 its clear that the own bus 3 is the most impacted one. This is however due to the unloading of the generators in the network, mostly by the synchronous condenser in bus 3.

The positive correlation of bus loading and short circuit capacity shown in this section cannot be generalized as the short circuit current are topology dependent.

Table 4.12: Numerical values of per-bus short-circuit current, large load disconnection.

Bus	Reference (p.u.)	Simulated (p.u.)	Δ (%)
1	30.45	30.40	-0.16
2	20.91	20.73	-0.87
3	8.60	8.25	-3.97
4	11.85	11.75	-0.78
5	12.23	12.17	-0.47
6	5.72	5.70	-0.37
7	6.07	6.04	-0.44
8	4.16	4.15	-0.29
9	5.48	5.46	-0.36
10	4.43	4.42	-0.23
11	3.94	3.93	-0.18
12	3.15	3.15	-0.13
13	4.10	4.09	-0.20
14	3.20	3.20	-0.10

4.2.4 Transmission line outage

Lastly, the short circuit current on the buses are to be evaluated in case of changes in the network topology. In this section, the line between buses 4 and 5 from Figure 4.1 are disconnected. This scenario is called **Simulated** in this present section.

From Table 4.13 is seen that the loading is kept constant but the unavailable path between buses 4-5 induces higher loses in the network due to power flow redirection.

In Figure 4.5 and Table 4.14 the impact of removing the line between buses 4 and 5 can be clearly seen in the short circuit currents in those same buses.

From the traditional short circuit calculation Equation 2.5, the previous change in the short circuit capacity in buses 4 and 5 can be explained by the change of the equivalent seen by these buses. This further reinforces the entangled behavior between network topology and short circuit capacity.

Table 4.13: Pre-fault condition for default operation point and transmission line 4-5 outage.

Metric	Reference	Simulated	Δ (%)
Total generation (MW)	272.39	275.10	+0.99
Total renewable (MW)	0.00	0.00	
Renewable share (%)	0.00	0.00	
Total load (MW)	259.00	259.00	+0.00
Losses (%)	4.92	5.85	+19.02

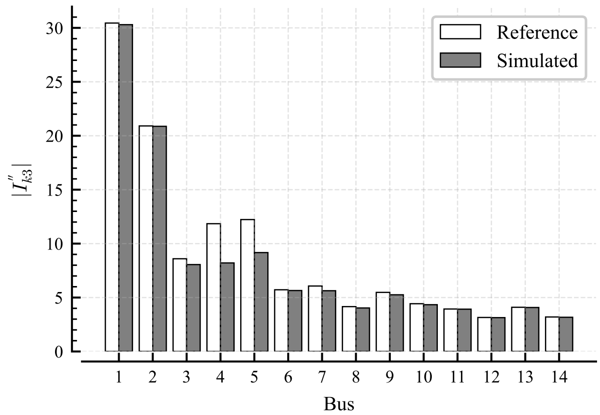


Figure 4.5: Per-bus short-circuit current, transmission line 4-5 outage.

Table 4.14: Numerical values of per-bus short-circuit current, transmission line 4-5 outage.

Bus	Reference (p.u.)	Simulated (p.u.)	Δ (%)
1	30.45	30.30	-0.49
2	20.91	20.87	-0.18
3	8.60	8.06	-6.27
4	11.85	8.21	-30.70
5	12.23	9.17	-25.02
6	5.72	5.65	-1.22
7	6.07	5.63	-7.18
8	4.16	4.03	-2.98
9	5.48	5.26	-4.05
10	4.43	4.34	-2.06
11	3.94	3.92	-0.37
12	3.15	3.14	-0.46
13	4.10	4.08	-0.48
14	3.20	3.17	-0.84

In this section, changes were made in the loads and topology with the generators responding to this change. The next section, changes will be actively made on the generators and the impacts on the short circuit capacity measured.

4.3 Impact of Generation Mix on Short-Circuit Capacity

The capability of the proposed method in handling active changes in the generation facilities is performed in this section. The changes comprise reduction in synchronous generator units, disconnection of synchronous generator units and a scenario without synchronous generation units.

4.3.1 Modified reference network

To extend the flexibility of the network and evaluate the method performance, additional synchronous generator units are included in the network, these are shown in Table 4.15.

Table 4.15: Generators added to the reference network.

Generator	Bus	P (MW)	V (p.u.)
G12	12	6.00	1.045
G13	13	7.00	1.045
G14	14	8.00	1.045

Additionally, converter interfaced generation as discussed in Section 3.3.2.3, are also included in the network, however, initially disconnected. These are shown in Table 4.16.

Table 4.16: Full converters added to the reference network.

Generator	Bus	P (MW)	Q (MVAr)
FC2	2	9.00	4.36
FC12	12	5.40	2.62
FC13	13	6.30	3.05
FC14	14	7.20	3.49

The loads are unchanged and total $S_{load} = 259.0 + j73.5$ MVA.

For the results to follow inside this Section 4.3, the **Reference** case is the modified reference network as defined before, i.e. all the previous elements of the reference network plus the additional generators of Table 4.15. As remarked previously, the converters from Table 4.16 are initially disconnected and thus does not contribute to the results of the **Reference** case. This definition of the reference case is used to evaluate the impacts of full converters in the short circuit calculation.

4.3.2 Effect of increasing full converter penetration

In this scenario full converters are connected to the grid and the generation from synchronous generators are reduced proportionally while supplying constant loading.

Generators at buses 2, 12, 13 and 14 have their active power setpoint reduced in the magnitude of the active power of the converters from Table 4.16. As observed in Table 4.17 the power flow is intentionally netted at the bus. The converters' contribution, $P_{conv} = 27.90$ MW represents 10.32% of the total network generation of $P = 270.32$ MW.

Table 4.17: Pre-fault condition for reference case and increased converter contribution.

Metric	Reference	Simulated	Δ (%)
Total generation (MW)	270.32	270.32	+0.00
Total renewable (MW)	0.00	27.90	
Renewable share (%)	0.00	10.32	
Total load (MW)	259.00	259.00	+0.00
Losses (%)	4.19	4.19	+0.00

In Table 4.18 it can be seen that increasing the contribution from full converters, while maintaining the synchronous generators still connected will increase the short circuit capacity on the buses adjacent to the ones with synchronous generators. As previously stated, the contribution from the machine to the short circuit comes mainly from its impedance, therefore, most of the contribution to the short circuit are still present inherently. On top of that, the converter contribution is also added.

Table 4.18: Numerical values of per-bus short-circuit current, increasing full converter penetration.

Bus	Reference (p.u.)	Simulated (p.u.)	Δ (%)
1	31.28	31.34	+0.21
2	22.02	22.00	-0.11
3	8.92	8.95	+0.35
4	13.56	13.63	+0.49
5	14.12	14.19	+0.47
6	9.58	9.66	+0.88
7	7.20	7.24	+0.56
8	4.47	4.48	+0.29
9	7.25	7.30	+0.75
10	5.59	5.63	+0.72
11	5.04	5.08	+0.79
12	6.68	6.68	-0.02
13	8.75	8.76	+0.13
14	6.49	6.47	-0.34

On the generator/converter buses its observed a local decrement in the short circuit level since the converter contribution is zero when the terminal voltage reached a minimum value, as presented in Section 3.3.2.3. However, this cannot be generalized due to topological influence, and also since the magnitude of changes are in the margin of error encountered in the validation (Section 4.1.2).

When observing bus 13 from Table 4.18, its short circuit capacity increases. Although the active power was reduced, the machine still plays a role on reactive support, resulting on a higher apparent power point of operation when comparing to the reference case, shown in Table 4.19.

Table 4.19: Generators loading for an increasing full converter penetration scenario.

Bus	Reference S (MVA)	Simulated S (MVA)	Δ_S (%)
1	209.77	209.77	+0.00
2	54.43	44.95	-17.41
3	23.38	23.38	+0.00
6	24.00	24.00	+0.00
8	16.70	16.70	+0.00
12	11.33	12.24	+8.04
13	11.73	12.48	+6.42
14	8.84	0.84	-90.46

4.3.3 Effect of synchronous generators phase-out

In this scenario the converters are turned online, but the synchronous machines are completely disconnected from the system. Generators of buses 2, 12, 13 and 14 are now offline, their contribution were 61.0 MW. Since now the power flow effect is not netted on the bus (note the differences of active power from Table 4.15 and Table 4.16), additional power is requested from the slack bus (Generator 1) to supply the additional losses. This is shown in Table 4.20.

Table 4.20: Pre-fault condition for reference case and synchronous generators phase-out.

Metric	Reference	Simulated	Δ (%)
Total generation (MW)	270.32	272.06	+0.64
Total renewable (MW)	0.00	27.90	
Renewable share (%)	0.00	10.26	
Total load (MW)	259.00	259.00	+0.00
Losses (%)	4.19	4.80	+14.65

In Figure 4.6 is clear the impact of the synchronous generators disconnection in the network short circuit capacity, with a significant reduction across all buses.

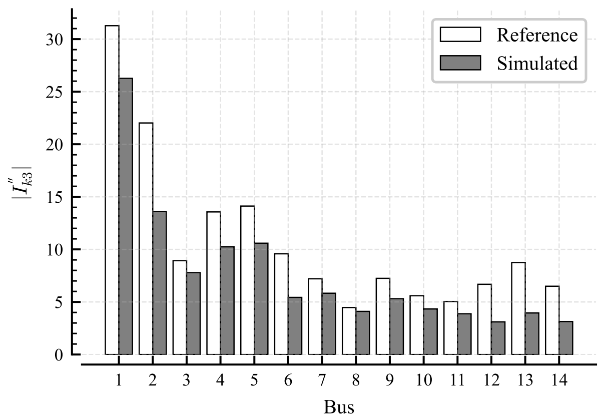


Figure 4.6: Per-bus short-circuit current, synchronous generators phase-out.

At Table 4.21 and Figure 4.7 is observed a higher relative impact on buses that used to have generators connected. This is expected, since the higher contribution of short circuit from the machine is when the short circuit is at its terminals, as seen in Figure 3.11.

Table 4.21: Numerical values of per-bus short-circuit current, synchronous generators phase-out.

Bus	Reference (p.u.)	Simulated (p.u.)	Δ (%)
1	31.28	26.27	-16.01
2	22.02	13.61	-38.20
3	8.92	7.79	-12.71
4	13.56	10.24	-24.48
5	14.12	10.59	-25.01
6	9.58	5.43	-43.26
7	7.20	5.83	-19.04
8	4.47	4.10	-8.17
9	7.25	5.31	-26.77
10	5.59	4.33	-22.48
11	5.04	3.87	-23.21
12	6.68	3.10	-53.55
13	8.75	3.95	-54.82
14	6.49	3.14	-51.70

From Figure 4.7 it can be seen that electrically distant buses are less affected, for example of bus 3 and 8.

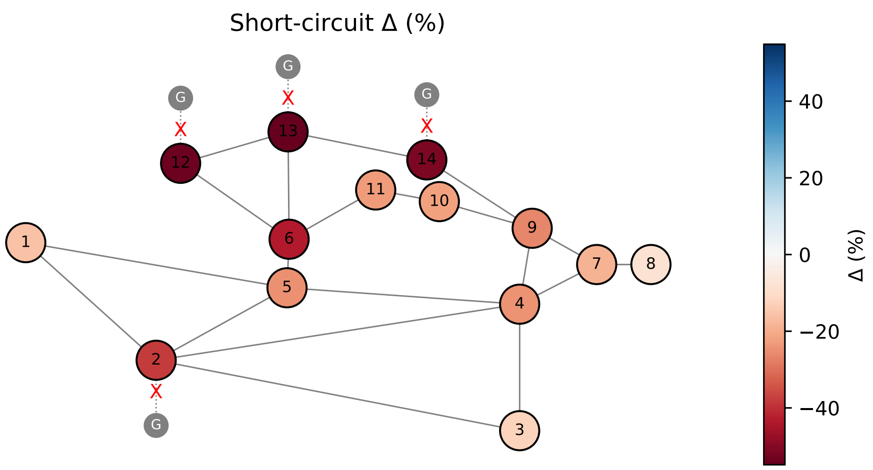


Figure 4.7: Spatial short-circuit current variation, synchronous generators phase-out.

Differently from generators decreasing their production where the short circuit capacity might increase or decrease, for generators phasing-out, the observed effect is exclusively of decreasing in the short circuit capacity of the network.

4.3.4 Converter-only generation

Lastly, in this section, all synchronous generators are turned off and the short circuit calculation is conducted and the response of the proposed method evaluated.

To set up this scenario, all converters from Table 4.16 are connected, and all the synchronous machines (Table 4.1 and Table 4.15) are disconnected. The slack generator from bus 1 is converted to an ideal current source ($Y_{Norton} = 0$). As it can be investigated in the Synchronous machine model Equation 3.19 and Equation 3.20, or even on the External

Grid model, Equation 3.35 and Equation 3.36, the slack generator model is then reduced to a current source with magnitude equal to the pre-fault current.

Most of the production comes from the slack bus, which can be seen as an ideal external network, the loading are kept constant as shown in Table 4.22.

Table 4.22: Pre-fault condition for reference case and converter-only generation.

Metric	Reference	Simulated	Δ (%)
Total generation (MW)	270.32	273.70	+1.25
Total renewable (MW)	0.00	27.90	
Renewable share (%)	0.00	10.19	
Total load (MW)	259.00	259.00	+0.00
Losses (%)	4.19	5.37	+28.28

The short circuit capacity of the buses, as show in Figure 4.8, collapses and the values plateau near the pre-fault current.

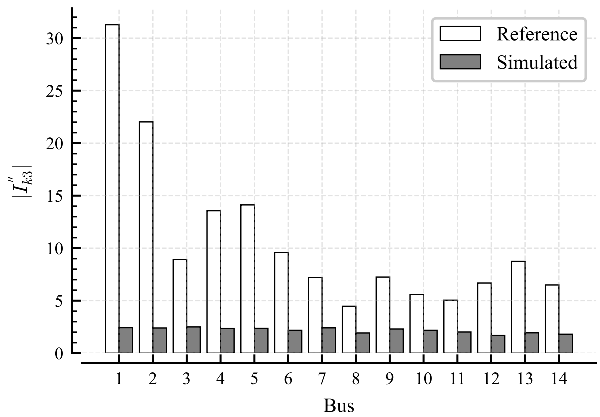


Figure 4.8: Per-bus short-circuit current, converter-only generation.

As the converters were operating on their rated capacity previous to the fault, and also do not contribute for short-circuits near their terminals, the converters additional contribution to the short-circuit is small. Since there is approximately a single current source at bus 1, Table 4.23 shows that the short circuit capacity decreases as it gets further from bus 1 (increasing electrical distance, e.g. impedance), similar behavior can be seen in single-source radial networks.

Table 4.24 shows that the pre-fault load on the slack bus is 2.57 p.u., and the target voltage is 1.06 p.u. (Table 4.1), thus the pre fault current is

$$I_{terminal} = \frac{S_m^{pf}}{V_m^{pf}} = \frac{2.57}{1.06} = 2.43 \text{ p.u.}$$

Which is the short circuit capacity of bus 1 from Table 4.23.

The short circuit calculation under different conditions was so far accessed changing electrical parameters, i.e. elements operational condition and network topology. The results so far were used to check the consistency of the modeling to the theory and expected electrical behavior considering the complete set of parameters required by the proposed method. In the next section, the proposed method is evaluated using reduced set of parameters.

Table 4.23: Numerical values of per-bus short-circuit current, converter-only generation.

Bus	Reference (p.u.)	Simulated (p.u.)	Δ (%)
1	31.28	2.43	-92.24
2	22.02	2.40	-89.11
3	8.92	2.50	-71.99
4	13.56	2.36	-82.59
5	14.12	2.37	-83.24
6	9.58	2.18	-77.24
7	7.20	2.41	-66.54
8	4.47	1.92	-56.98
9	7.25	2.30	-68.26
10	5.59	2.18	-60.98
11	5.04	2.02	-60.03
12	6.68	1.70	-74.61
13	8.75	1.94	-77.85
14	6.49	1.81	-72.20

Table 4.24: Generators loading for a converter-only generation scenario.

Bus	Reference S (MVA)	Simulated S (MVA)	Δ_S (%)
1	209.77	257.12	+22.57
2	54.43	0.00	-100.00
3	23.38	0.00	-100.00
6	24.00	0.00	-100.00
8	16.70	0.00	-100.00
12	11.33	0.00	-100.00
13	11.73	0.00	-100.00
14	8.84	0.00	-100.00

4.4 Sensitivity Analysis and Error Quantification

To extend the proposed method towards practical implementation in real electrical power system, this section aims to perform analysis and quantify the discrepancies encountered when some parameters required by the method is not available and therefore need to be approximated. The network used in this section is the modified reference network as described in Section 4.3.1.

Referring to Figure 3.17, the approach used in this section is to use the method similarly as before but then override the parameter of interest just before the short circuit calculation. In other words, the power flow step is conducted with the integral network without overriding any values, in order to provide realistic pre-fault conditions, as these are measured in real power systems.

The reference value of short circuit calculation, **Reference** case, considers all generators and converters from Table 4.15 and Table 4.16 to be connected to the network. The total amount of generation, load and losses are constant between the reference case and the simulated case.

4.4.1 Effect of neglecting loads on short-circuit capacity

Loads are often neglected in transmission level short circuit calculations [48]. Conversely, this proposed method includes loads as constant impedance elements, as described in Section 3.3.2.1.

Table 4.25: Numerical values of per-bus short-circuit current, neglecting loads.

Bus	Reference (p.u.)	Simulated (p.u.)	Δ (%)
1	31.34	31.67	+1.05
2	22.00	22.31	+1.39
3	8.95	9.09	+1.54
4	13.66	13.97	+2.33
5	14.22	14.56	+2.39
6	9.64	9.84	+2.06
7	7.24	7.43	+2.62
8	4.48	4.62	+3.14
9	7.29	7.45	+2.21
10	5.61	5.81	+3.45
11	5.07	5.27	+3.99
12	6.67	6.80	+1.87
13	8.75	8.92	+1.96
14	6.47	6.62	+2.24

From the data acquired from the simulation show in Table 4.25, it can be seen that if neglecting loads modeling in the short circuit calculation, one might be assuming a maximum error of 4.0%. Table 4.25 also can be interpreted as the marginal contribution of loads to short-circuit currents.

Besides its model, several effects influence the impact of loads in the short circuit current, between them are the equivalent impedance of the system and the impedance of the load. From Equation 3.6, the equivalent impedance can assume values that can make the calculated short circuit current larger or smaller than the reference value. Network topology further introduces variables that make the load-neglecting behavior hard to generalize, however it is seen in this section that load-neglecting impact are several times smaller and more uniform across the network than a generator disconnection from Section 4.3.3.

4.4.2 Effect of neglecting converter contributions' on short-circuit capacity

The converters from Table 4.16 are quite small and their impact on the short circuit current would thus be small. For this section, the converter of bus 12 are replaced for one with $P_{rated} = 400$ MW and operating with pre-fault condition of $P_m^{pf} = 10$ MW. As one might conclude from the converter model from Section 3.3.2.3, a converter operating near its rated power does not inject much additional short-circuit current due to saturation (Equation 3.28).

In Table 4.26 its observed that the most affected buses are buses 6 and 13, while the converter bus, bus 12, have small change. As before, in the case of loads at Section 4.4.1, Table 4.26 can be read as the marginal contribution of converters to short circuit current.

Table 4.26: Numerical values of per-bus short-circuit current, neglecting converters.

Bus	Reference (p.u.)	Simulated (p.u.)	Δ (%)
1	31.48	31.22	-0.83
2	22.19	21.94	-1.15
3	9.01	8.89	-1.27
4	13.97	13.53	-3.16
5	14.64	14.09	-3.74
6	11.41	9.49	-16.83
7	7.37	7.16	-2.86
8	4.51	4.45	-1.42
9	7.48	7.19	-3.96
10	5.78	5.54	-4.17
11	5.34	5.00	-6.36
12	6.67	6.62	-0.86
13	10.31	8.66	-15.97
14	6.66	6.43	-3.51

Since the converter in bus 12 is at least 40 times the size of the second-largest converter, most of the converter contribution to short circuit in this network comes from it. Therefore, in Figure 4.9 is evident that converter contribution immediately impact the neighboring buses, and this impact decreases with as the electrical distance increases. Buses in the extremity from bus 12 are very little affected. This effect is encoded on the full converter model by the Fault Ride-Through as described in [18, 19].

Although this converter has active power at least 50 times larger than the synchronous generators from Table 4.15, its contribution to short circuit is significantly smaller, as seen in Table 4.21. Taking as example bus 6, the converter contribution ($P_{rated} = 400$ MW) at Table 4.26 is 1.89 p.u., or 16% of the bus capacity, while the 3 synchronous generators ($P_{rated} = 21$ MW) contribution from Table 4.21 sums up to approximately 4.15 p.u., or 43% of the bus capacity.

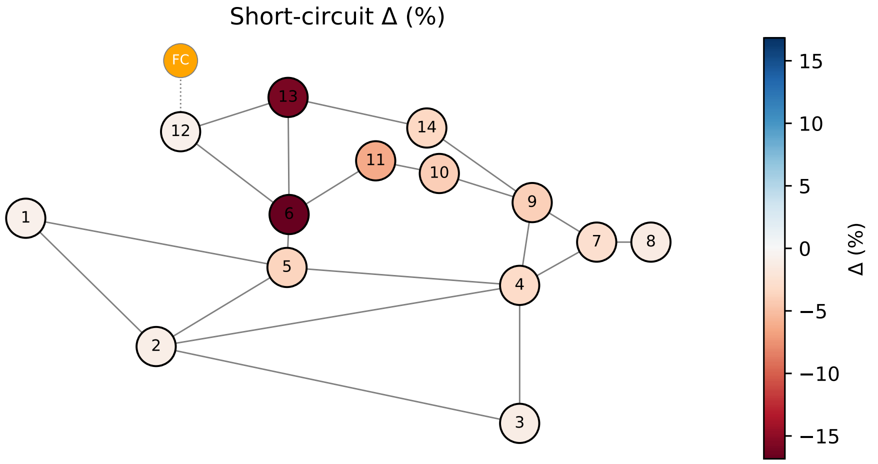


Figure 4.9: Spatial short-circuit current variation, neglecting converters. The 400 MW converter is shown in bus 12.

4.4.3 Effect of neglecting external grid contribution to short-circuit capacity

For this scenario, an external grid connection at the voltage level $V = 138$ kV and with short circuit capacity $S_{SC} = 1000$ MVA is added to bus 13, this is equivalent to a short circuit current of $I_{SC} = 4.2$ kA, or 10.0 p.u. in the 138kV/100MVA base, as it can be confirmed by the equations of Section 3.3.2.4. The external grid exchange no power to the current network.

In Figure 4.10 the bus which the external grid is connected, bus 13 present a clear change in short circuit capacity due to the omission of the external grid in the calculation. Conversely, the external grid short circuit current contribution is significant in its connection bus.

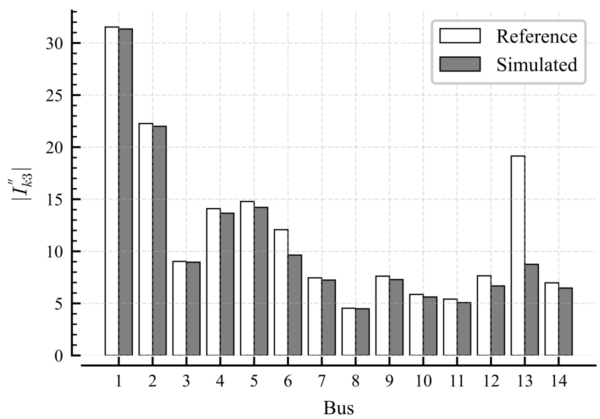


Figure 4.10: Per-bus short-circuit current, neglecting external grid.

As seen in Table 4.27, bus 13 neighboring buses (6, 12 and 14), have significant contribution from the external grid short circuit capacity, 12%, 20% and 7%, respectively. In bus 13, the external grid short circuit current contribution is approximately 10.4 p.u., then considering the pre-fault voltage of $V_m^{pf} = 1.044$ p.u. obtained in the power flow solution, the current value can be easily confirmed as the current will have a scaling factor proportional to the off-nominal voltage ($\frac{1.044}{1.0}$) since the external grid is considered to be operating without power exchange prior to the fault condition.

Table 4.27: Numerical values of per-bus short-circuit current, neglecting converters.

Bus	Reference (p.u.)	Simulated (p.u.)	Δ (%)
1	31.53	31.34	-0.60
2	22.27	22.00	-1.20
3	9.03	8.95	-0.80
4	14.10	13.66	-3.13
5	14.78	14.22	-3.82
6	12.08	9.64	-20.18
7	7.45	7.24	-2.85
8	4.53	4.48	-1.16
9	7.61	7.29	-4.30
10	5.86	5.61	-4.21
11	5.41	5.07	-6.29
12	7.65	6.67	-12.73
13	19.15	8.75	-54.31
14	6.97	6.47	-7.18

4.4.4 Effect of pre-fault complex voltage approximations on short-circuit capacity calculations

In this section, instead of acquiring the complex pre-fault voltage from the power flow solution (or from an SCADA system), the complex pre-fault voltages are approximated to pre-defined values. The **Reference** case consists of a short circuit calculation using the complex voltages from the power flow solution while the **Simulated** case is computed using pre-defined voltage magnitude and angle of $V = 1.05\angle 0.0°$ p.u..

The proposed method lacks the simplification of a flat voltage when compared to the traditional short circuit calculation using the impedance matrix. The impedance matrix method, as seen previously, the short circuit current in the bus is calculated directly using the diagonal element of the Z-Bus and the pre-fault voltage [21], therefore, the pre-fault complex power is encoded in the voltage and does not need explicit definition. However, in the proposed method, the pre-fault complex power is a parameter for the active elements. In this section, in addition to the pre-defined complex voltage (magnitude and angle), the elements will have a pre-fault complex power of $S = 0 + j0$ MVA. This means no current is injected by these elements prior to the fault.

The consequence of the complex power assumption on the active elements model is that loads will become open circuits, Equation 3.17, and generators will have no pre-fault current injection, Equation 3.18. Converters response changes with the pre-fault operation point, as observed in Figure 3.13, but without any distinguished behavior due to the complex power assumption. External grids are modeled similar to generators and the pre-fault complex power change its response during a short-circuit, however, as stated in the previous Section 4.4.3, the external grids are already assumed to be unloaded prior to the short-circuit.

As in Section 4.4.1, the **Reference** case is the modified reference network with the additional generators (Table 4.15) and converters (Table 4.16) connected. Table 4.28 presents the results for the simulation.

Table 4.28: Numerical values of per-bus short-circuit current, pre-defined complex voltage $V = 1.05\angle 0.0°$ p.u..

Bus	Reference (p.u.)	Simulated (p.u.)	Δ (%)
1	31.34	31.12	-0.71
2	22.00	21.92	-0.34
3	8.95	8.92	-0.37
4	13.66	13.77	+0.83
5	14.22	14.36	+1.03
6	9.64	9.69	+0.53
7	7.24	7.21	-0.37
8	4.48	4.39	-2.02
9	7.29	7.28	-0.12
10	5.61	5.67	+1.05
11	5.07	5.16	+1.71
12	6.67	6.88	+3.14
13	8.75	8.98	+2.66
14	6.47	6.62	+2.22

As seen on previous chapters and for this particular network, the short circuit contribution comes mainly from synchronous generators, additionally their contribution are strongly related to their reactances (and less to their pre-fault loading), therefore as seen on Table 4.28, the error on the buses short circuit current is maximum on bus 8, 7%. This is because of the difference between bus 8 true voltage value of $V = 1.09\angle -10.01°$ p.u.. Bus 3 has similar difference in voltages, but its percent error is masked by its higher short circuit level.

In Figure 4.11, a sensitivity of the buses absolute difference in per-unit is presented for different pre-defined bus voltage magnitude. The y-axis shows the difference between the calculated short circuit current using the true value (power flow solution) and the approximated fixed pre-fault value. It is observed that the fixed pre-fault voltage value choice have an asymmetrical effect on buses error, as it can be observed first by the increasing absolute value of the median and secondly by the increasing variance, as the pre-fault voltage moves away from 1.05 p.u..

For this network, the average bus voltage is between 1.05 and 1.10 p.u., as seen in Section 4.1.1, it is also observed in Figure 4.11 that the median of the differences of the cases $V = 1.05$ and $V = 1.10$ are closer to zero with smaller spread on the data. The cases $V = 0.95$ and $V = 1.00$ and show significant outliers of approximately $\Delta I_{SC} = -3.2$ p.u. and $\Delta I_{SC} = -1.7$ p.u., respectively. This means that the pre-defined value of choice will define the accuracy of the calculations, although typical values are used for short-circuit calculations, the true value will depend on the specific network topology and loading.

4.5 Spatial Analysis of Short-Circuit Capacity

This section will summarize the scenarios presented in Sections 4.2, 4.3, and 4.4. Figure 4.12 present each topic on the y-axis, the bus on the x-axis and each cell is the difference from the reference case, as outlined in the previous sections for the 14-bus test case.

It can be seen that the variable that most strongly drives the short circuit capacity of the network is the presence and connection of synchronous generators, since the scenarios with the largest changes are synchronous generators phase-out (Section 4.3.3) and no synchronous

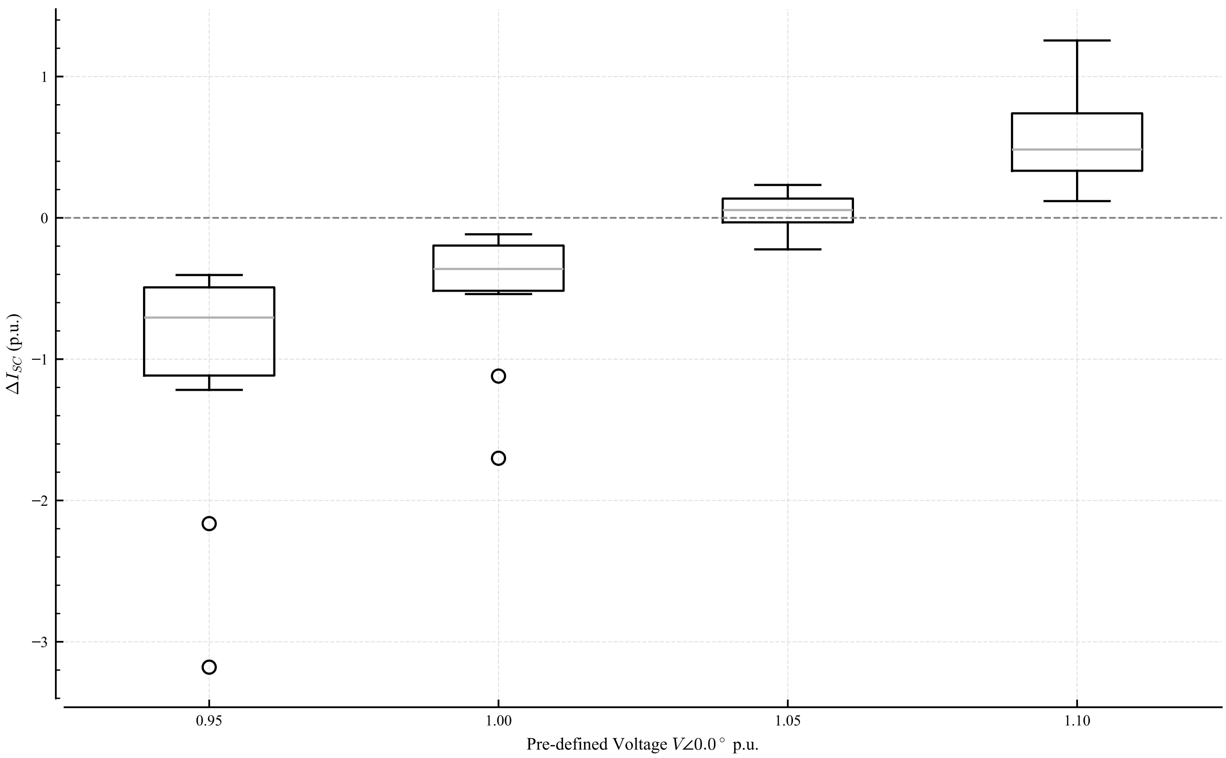


Figure 4.11: Buses I_{SCC} deviation from reference for a given pre-defined voltage magnitude.

generators (Section 4.3.4), i.e. disconnecting generators from the system. It is also observed the local effect of topology change (Section 4.2.4), reduction of short circuit capacity of buses 4 and 5 when disconnecting the line between 4 and 5. Full size converters although contribute to short circuit, normally up to 1.2 times their rated current, have modest contribution, even when compared to synchronous machine a fraction of their power. Additionally, external grids are an important factor and need to be included in the best extent to achieve accurate results. Lastly, the other scenarios presented do have influence in the short circuit capacity but in a minor extension.

4.6 Temporal and Continuous Assessment

As discussed previously on Section 2.1 and in the respective models in Section 3.3, the proposed methodology solution is a complex number (phasor) representing the short circuit current/power for each bus in the network, and not a function of time such as in electromagnetic transient studies. Each solution is called a snapshot and captures the state for a singular condition in time. This work then proposed the temporal assessment of the short circuit capacity by stacking individual snapshots in an ordered way. This is a reasonable assumption when comparing the timescale under study of $40 \sim 50$ ms, as discussed in Section 3.3.2.3, and the supervisory system sampling interval of ~ 1 s [35].

A daily time series is composed by creating 24 individual operating points to the modified reference network. This is achieved by creating a background load variation and including a controlled operation of production facilities on top. The load shape is modeled as a Markov chain with 5 load level, including seasonality and weekday/weekends distinction, using historical load data (between January/2016 and October/2025) from the Swedish SE2

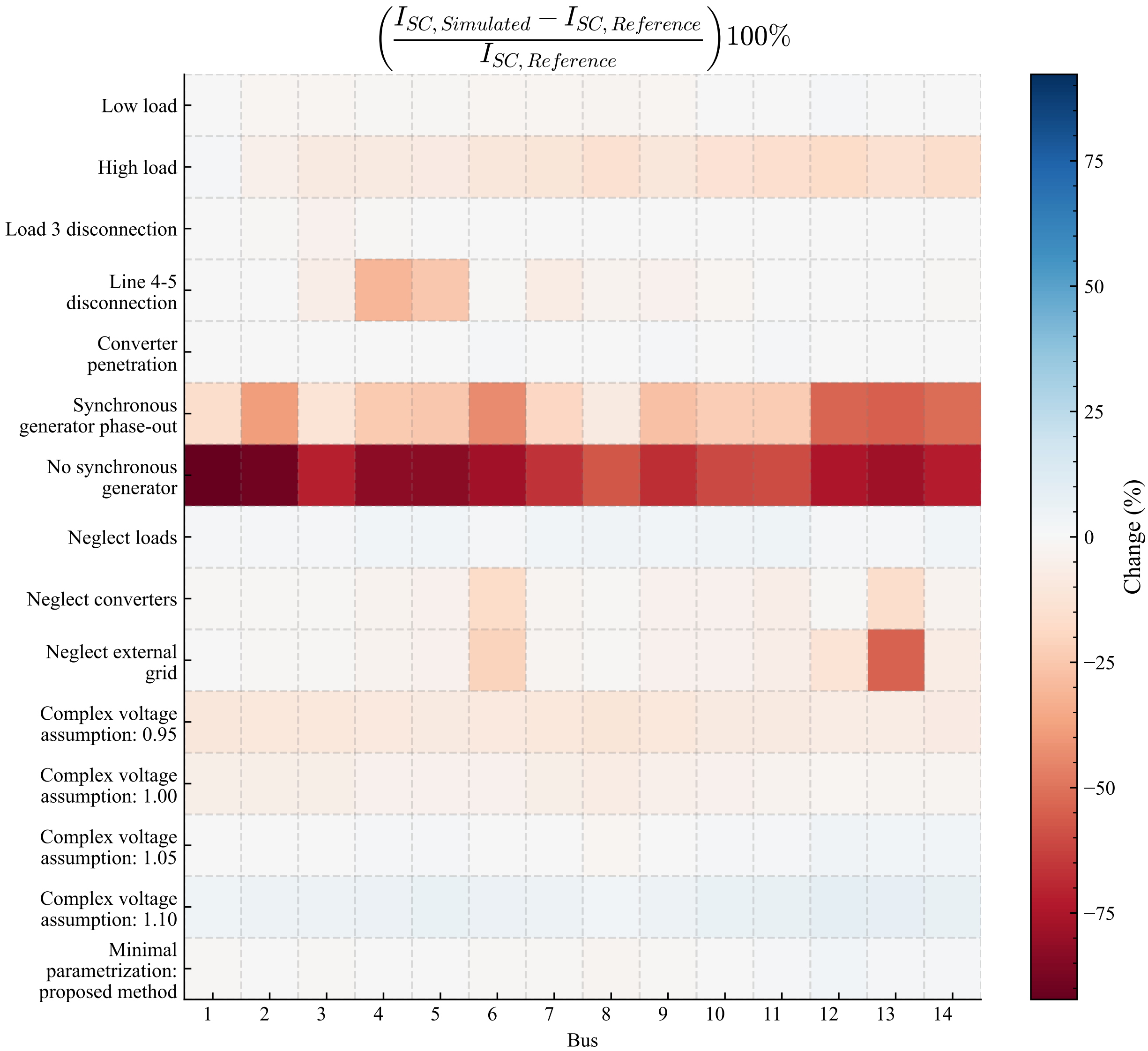


Figure 4.12: Heatmap for scenarios. Cell values represent the percent difference of the Simulated case to the Reference case. Negative values indicate that the simulated case is lower than the reference case.

bidding zone [64], then a weekday in winter is chosen. The sampled load value is then normalized by the historical median (between quantiles 0.1% and 99.9%) and the scaling factor applied to the loads of the modified reference network (Table 4.2), each load having its own random sampling.

For generation, Between 9 a.m. and 15 p.m. all full converters (Table 4.16) are turned on and are operating in their rated capacity. The synchronous machines follow the sequence of operation as presented in Table 4.29. The generators initially reduce their production, then goes offline, later they reconnect and ramp up their power again.

Table 4.29: Synchronous generators coordinated sequence of operation. The synchronous condensers at buses 3, 6 and 8 are always online.

Time	Synchronous Generator	Obs
9 a.m.	G1, G2, G12, G13, G14	Reduced generation
10 a.m.	G1, G2	Reduced generation + generators turned off
11 a.m.	G1, G2	no change
12 p.m.	G1	Generator turned off
13 p.m.	G1, G2	Generator turned on
14 p.m.	G1, G2	no change
15 p.m.	G1, G2, G12, G13, G14	Generators turned on

Figure 4.13 present the resulting timeseries for buses 2, 5, 8 and 14. It is noticed that around 10 a.m., when generators 12, 13 and 14 are turned off, the short circuit capacity is decreased across the network, and is further decreased when G2 is disconnected, specially noted in bus 2. However in bus 8, the generators turning on and off cause small changes in its capacity. Bus 14 also have a smaller change in short circuit capacity, when compared to bus 2 and 5, when generator of bus 2 is turned off, around 12 p.m..

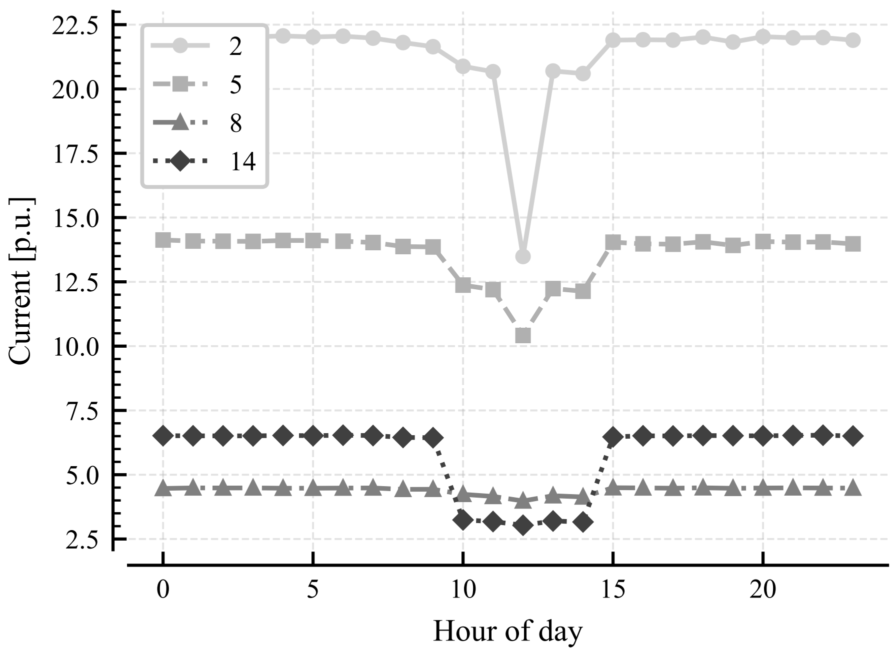


Figure 4.13: Timeseries for coordinated synchronous generator operation.

Afterward, a new operation sequence is composed by randomly choosing which generators are to be online, generator 1 however is set to be always online. The result is show in Figure 4.14.

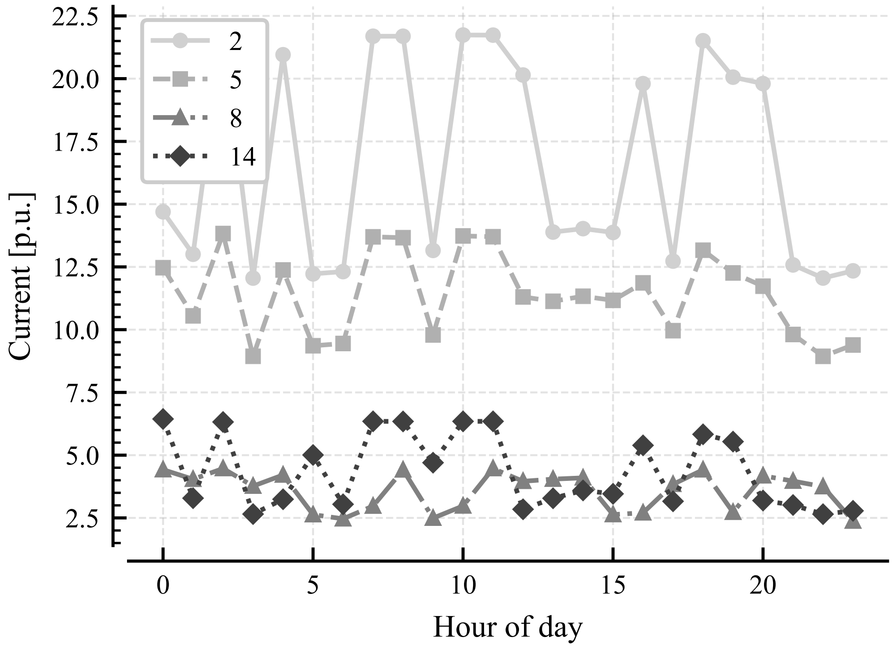


Figure 4.14: Random sequence timeseries.

The random sequence is used to access the resilience of the proposed method. It was found that given specific combinations of high system loading and few synchronous generators online, the short circuit calculation Algorithm 1 does not find a solution within a residual tolerance of 10^{-6}. This issue is direct consequence of the non-linear elements. Therefore, this leads to two main solutions, the first being implementing a more robust algorithm with explicit Jacobian computation (instead of numerically computation as of it is now), and the second to assume some loss of accuracy by neglecting or linearizing of nonlinear elements.

The next section explore a reduced-order model where the buses' voltage are assumptions and also nonlinear elements, within the scope of this work, such as converters and loads, are neglected. The latter reduces the model to a linear system of equations and thus avoiding the divergence issue.

4.7 Minimal Parametrization: Reduced-Order Model Implementation

Extending the discussion of the previous sections, the present combine those assumptions and simplifications to achieve an implementation using a reduced order model. This aims to encourage the implementation of continuous short circuit calculation on real power systems, even when some measurements or data are not available. External grids are not included in the **Reference** case or the **Simulated** case of this section, however, as shown previously in Section 4.4.3, its omission may lead to significant local impact. Therefore, they should be included to the fullest extent possible. No change is needed in the short circuit calculation algorithm as presented in Algorithm 1.

To implement the reduced order model the following information is needed: impedance for all passive elements (transmission lines, transformers and shunt elements), synchronous machines reactances and external grids short circuit capacity. The assumptions considered here are: voltages in all buses are set to $V = 1.05\angle 0.0°$ p.u. and all active elements are unloaded $S = 0+j0$ MVA. Additionally, these simplifications are made: loads and converters are neglected in the short circuit calculation. Table A.1 in the Appendix presents in detail the parameters used by the proposed method and reduced order model.

As in Section 4.4.1 and Section 4.4.4, the reference case is the modified reference network with the additional generators and converters connected. Table 4.30 presents the results for the simulation.

Table 4.30: Numerical values of per-bus short-circuit current, reduced-order model.

Bus	Reference (p.u.)	Simulated (p.u.)	Δ (%)
1	31.34	31.02	-1.03
2	22.00	21.89	-0.51
3	8.95	8.89	-0.76
4	13.66	13.66	+0.01
5	14.22	14.24	+0.20
6	9.64	9.56	-0.86
7	7.24	7.16	-1.10
8	4.48	4.38	-2.29
9	7.29	7.20	-1.16
10	5.61	5.62	+0.17
11	5.07	5.12	+0.88
12	6.67	6.83	+2.39
13	8.75	8.89	+1.66
14	6.47	6.59	+1.75

The largest error found in Table 4.30, 7.33%, have similar magnitude as found in Table 4.28, 7.08%, meaning that for this specific network, a significant part of the deviation comes from the voltage approximation and not the loads and converters contribution.

One might notice that the reduced order model, although modeled through the Norton equivalent and Admittance matrix, have similar parameters as the traditional short circuit calculation using the Thevenin equivalent and the Impedance matrix, since no non-linear element are now included. Therefore, Table 4.31 shows the differences between the short circuit calculation using the reduced order model (**Simulated**) and the **Traditional** method.

Table 4.31 rises important differences from both methods, specially how the previous assumptions affect their result. On the traditional method, the pre-defined voltage is applied directly to the short circuit calculation (Equation 2.5) and it represents the Thévenin equivalent seen from the faulted bus. On the proposed method, the pre-defined voltage is instead the initial condition of the active elements. This voltage assumption therefore acts on conceptually different values from both methods. Referring to the **Reference** case from Table 4.30, the traditional method is found to produce the smaller short circuit current values than the proposed method.

Other methods such as IEC 60909 [23], although similar to the traditional method, explicitly include the contribution of converters, performing the calculation for the network without them first to only them include later. This avoids masking the converters into the equivalent voltage seen from the faulted bus, when performing a pre-fault estimation using power flow, as it was done in Table 4.31, and consequently rising those differences.

Table 4.31: Numerical values of per-bus short-circuit current, comparison between proposed and traditional methods for the reduced-order model.

Bus	Simulated (p.u.)	Traditional (p.u.)	Δ (%)
1	31.02	30.90	-0.37
2	21.89	21.76	-0.58
3	8.89	8.82	-0.75
4	13.66	13.57	-0.61
5	14.24	14.21	-0.22
6	9.56	9.21	-3.60
7	7.16	6.89	-3.74
8	4.38	4.24	-3.10
9	7.20	6.87	-4.63
10	5.62	5.37	-4.45
11	5.12	4.91	-4.04
12	6.83	6.68	-2.33
13	8.89	8.66	-2.62
14	6.59	6.41	-2.71

Chapter 5

Conclusions

There are many methods to perform short circuit calculations in non-radial systems found in the literature, and as many different electrical components model. Each method aims to achieve the balance between accuracy and computational expenditure for specific studies, in general Electromagnetic Transient-based methods often provides more accurate results and Phasor-based methods provides less computational expensive. This work focused on slower power system dynamic phenomena (time scale $\geq$ 40 ms), thus, Phasor-based methods provides the best trade-off between accuracy and computational expenditure.

Power electronics devices offers flexible and fast operation on one hand and modelling complexity on other. Traditional short circuit calculation methods are suited for a network with only linear or linerized elements, it is therefore an additional effort to include power electronics elements in such calculations. A common approach to include power electronics/nonlinear elements, and the one used in this work, is to model these elements using the Norton equivalent with a controlled current source and then solve a $YV = I$ set of equations using an iterative method.

Similarly as the traditional method may fail numerically due to an ill-conditioned admittance matrix, iterative methods may as well present numerical instability issues due to the nonlinearities of the power electronics sources.

Results shown that the Short Circuit current/power is not static value, it varies with several parameters of the power system including the network loading and topology, but most critically with the amount of synchronous generator-based generation connected at a given moment. Converter-Interfaced Generation although provides Short Circuit current/power, it do in an order or magnitude much smaller then synchronous generators. The heighest impact in the Short Circuit capacity is when synchronous generators are disconnected from the system, conversely, the reduction of these machines active power output while remaining connected to the system did not presented such impact. This observation highlight the role of dedicated synchronous condensers and synchronous condenser operation mode of hydro power plants.

The method develop in this work was validated against commercial software with maximum error of 2.32%. The proposed method is flexible enough to include full size converters and other nonlinear elements while the reduced order model is designed to specific necessities or reduced information availability, specicially PMU measurements and full converter control details. The reduced order model showed a maximum error of 2.39%.

Finally, the main findings are that the Short-Circuit Calculation is very sensitive to synchronous generators, so even simplified methods for continous short circuit calculation methods are of great values since it gives insight and acts as a screening method and provide possible directions for detailed studies. Increasing in Converter-Interfaced Generation, topology changes, planned maintenance as seen in this work all affects the Short Circuit capacity throughout the network, therefore, relay protection configuration must be awere of such changes to still fulfill its intendend functionality.

Bibliography

[1] Svenska Kraftnät. *Power system operational security - future needs and abilities.* Tech. rep. Aug. 2025. URL: https://www.svk.se/4ae285/siteassets/om-oss/rapporter/2025/power-system-perational-security.pdf.

[2] FINGRID. *Inertia of the Nordic power system.* Feb. 2026. URL: https://www.fingrid.fi/en/electricity-market-information/InertiaofNordicpowersystem/.

[3] International Energy Agency. *World Energy Outlook 2025.* 2025. URL: https://www.iea.org/reports/world-energy-outlook-2025.

[4] FINGRID. "Utilizing Equivalent Short-Circuit Ratio (ESCR) approach for assessing the slow converter driven stability and tuning the voltage controllers". Feb. 2023. URL: https://www.fingrid.fi/globalassets/dokumentit/fi/palvelut/kulutuksen-ja-tuotannon-liittaminen-kantaverkkoon/guidance_for_escr_and_voltage_control-1.pdf.

[5] Nikos Hatziargyriou et al. "Definition and Classification of Power System Stability – Revisited & Extended". In: *IEEE Transactions on Power Systems* 36.4 (2021), pp. 3271–3281. DOI: 10.1109/TPWRS.2020.3041774.

[6] Paul Denholm et al. *Inertia and the Power Grid: A Guide Without the Spin.* Tech. rep. NREL/TP-6A20-73856. National Renewable Energy Laboratory, 2020. URL: https://www.nrel.gov/docs/fy20osti/73856.pdf.

[7] Hongchun SHU et al. "Estimation of inertia demand in power systems considering wind power virtual inertia". In: *International Journal of Electrical Power & Energy Systems* 167 (2025), p. 110597. ISSN: 0142-0615. DOI: https://doi.org/10.1016/j.ijepes.2025.110597. URL: https://www.sciencedirect.com/science/article/pii/S0142061525001486.

[8] E Muljadi et al. "Understanding inertial and frequency response of wind power plants". In: *2012 IEEE Power Electronics and Machines in Wind Applications.* 2012, pp. 1–8. DOI: 10.1109/PEMWA.2012.6316361.

[9] Tianshu Bi et al. "Review on renewable energy source fault characteristics analysis". In: *CSEE Journal of Power and Energy Systems* 8 (4 2022), pp. 963–972. ISSN: 2096-0042.

[10] Muntathir Al Talaq, Mohammad AlMuhaini, and Hamed Bin Qadhi. "Renewable Energy Sources Contribution to Short Circuit Current". In: *2023 IEEE PES 15th Asia-Pacific Power and Energy Engineering Conference (APPEEC).* 2023, pp. 1–5. DOI: 10.1109/APPEEC57400.2023.10561996.

[11] Murillo Cobe Vargas, Oureste Elias Batista, and Yongheng Yang. "Estimation Method of Short-Circuit Current Contribution of Inverter-Based Resources for Symmetrical Faults". In: *Energies* 16 (7 2023). ISSN: 1996-1073. DOI: 10.3390/en16073130. URL: https://www.mdpi.com/1996-1073/16/7/3130.

[12] August Haväng and Simon Klangeryd. "Short-Circuit Capacity from Inverter-Based Resources". MA thesis. Lund University, June 2025.

[13] Sandra Thengius. "Fault current injection from power electronic interfaced devices". MA thesis. KTH Royal Institute of Technology, Sept. 2020.

[14] Paul Denholm and Ben Kroposki. *Understanding Power Systems Protection in the Clean Energy Future*. Tech. rep. NREL/TP-6A40-82269. National Renewable Energy Laboratory, 2022. URL: https://docs.nrel.gov/docs/fy22osti/82269.pdf.

[15] North American Electric Reliability Corporation. *Short-Circuit Modeling and System Strength*. Tech. rep. NERC, Feb. 2018. URL: https://www.nerc.com/globalassets/programs/rapa/ra/short__circuit__whitepaper__final__1__26__18.pdf.

[16] Zheng Xu et al. "The Definition of Power Grid Strength and Its Calculation Methods for Power Systems with High Proportion Nonsynchronous-Machine Sources". In: *Energies* 16 (4 2023). ISSN: 1996-1073. DOI: 10.3390/en16041842. URL: https://www.mdpi.com/1996-1073/16/4/1842.

[17] Aleksandar Boričić and Marjan Popov. "Voltage vulnerability curves: Data-driven dynamic security assessment of voltage stability and system strength in modern power systems". In: *International Journal of Electrical Power & Energy Systems* 168 (2025), p. 110636. ISSN: 0142-0615. DOI: https://doi.org/10.1016/j.ijepes.2025.110636. URL: https://www.sciencedirect.com/science/article/pii/S0142061525001875.

[18] Energimarknadsinspektionen. *Krav på spänningskvalitet i ett föränderligt elsystem*. Tech. rep. PP331119, Rev. 2. Feb. 2021. URL: https://ei.se/download/18.1106b27c18093d411c21211/1651837080803/Konsultrapport-Krav-p%C3%A5-sp%C3%A4nningskvalitet-i-ett-f%C3%B6r%C3%A4nderligt-elsystem-DNV-2021.pdf.

[19] *Kommissionens förordning om fastställande av nätföreskrifter med krav för nätanslutning av generatorer, (EU) 2016/631RfG*. 2016. URL: https://eur-lex.europa.eu/eli/reg/2016/631/oj?locale=sv#.

[20] Nasser D Tleis. "8 - Network equivalents and practical short-circuit current assessments in large-scale ac power systems". In: *Power Systems Modelling and Fault Analysis*. Ed. by Nasser D Tleis. Newnes, 2008, pp. 485–519. ISBN: 978-0-7506-8074-5. DOI: https://doi.org/10.1016/B978-075068074-5.50012-6. URL: https://www.sciencedirect.com/science/article/pii/B9780750680745500126.

[21] J. Duncan Glover, Thomas J. Overbye, and Mulukutla S. Sarma. *Power system analysis & design*. eng. Sixth edition. Boston, MA: Cengage Learning, 2017. ISBN: 1305632133.

[22] Federico Milano. *Power System Modelling and Scripting*. 1st ed. Springer Nature, 2010. ISBN: 3642136699.

[23] *IEC 60909-0: Short-circuit currents in three-phase a.c. systems - Part 0: Calculation of currents*. 2016.

[24] J C Das. *Power system analysis : short-circuit load flow and harmonics*. Second edition. CRC Press, 2012. ISBN: 1-315-21794-5.

[25] Luka Strezoski, Marija Prica, and Kenneth A Loparo. "Generalized Δ-Circuit Concept for Integration of Distributed Generators in Online Short-Circuit Calculations". In: *IEEE Transactions on Power Systems* 32 (4 2017), pp. 3237–3245. DOI: 10.1109/TPWRS.2016.2617158.

[26] Luka V STREZOSKI et al. “Modeling challenges and potential solutions for integration of emerging DERs in DMS applications: power flow and short-circuit analysis”. In: *Journal of Modern Power Systems and Clean Energy* 7 (6 2019), pp. 1365–1384. ISSN: 2196-5420. DOI: 10.1007/s40565-018-0494-1. URL: https://doi.org/10.1007/s40565-018-0494-1.

[27] Leon Thurner and Martin Braun. “Vectorized Calculation of Short Circuit Currents Considering Distributed Generation - An Open Source Implementation of IEC 60909”. In: (Feb. 2018).

[28] Deepak Deepak et al. “Load-Flow-Based Calculation of Initial Short-Circuit Currents for Converter-Based Power System”. In: *Energies* 18 (15 2025). ISSN: 1996-1073. DOI: 10.3390/en18154045. URL: https://www.mdpi.com/1996-1073/18/15/4045.

[29] Jie Song et al. “A short-circuit calculation solver for power systems with power electronics converters”. In: *International Journal of Electrical Power & Energy Systems* 157 (2024), p. 109839. ISSN: 0142-0615. DOI: https://doi.org/10.1016/j.ijepes.2024.109839. URL: https://www.sciencedirect.com/science/article/pii/S0142061524000607.

[30] S Tu et al. “Practical Calculation for Short Circuit Current in Power System without Synchronous Generator Source”. In: *2025 7th International Conference on Information Science, Electrical and Automation Engineering (ISEAE)*. 2025, pp. 59–64. DOI: 10.1109/ISEAE64934.2025.11041859.

[31] Wang Hongfu et al. “A Weighted Average Iteration Method for Improving Convergence of Short-Circuit Current Calculation with Controlled Current Source”. In: *2023 5th Asia Energy and Electrical Engineering Symposium (AEEES)*. 2023, pp. 897–903. DOI: 10.1109/AEEES56888.2023.10114227.

[32] Jian Wang et al. “Screening of Areas with High Risk of Short-Circuit Current Exceedance in Power Systems with Renewable Energy Integration”. In: *2024 6th Asia Energy and Electrical Engineering Symposium (AEEES)*. 2024, pp. 396–401. DOI: 10.1109/AEEES61147.2024.10544651.

[33] Ye Yuan et al. “Inverse Power Flow Problem”. In: *IEEE transactions on control of network systems* 10 (1 2023), pp. 261–273. ISSN: 2325-5870.

[34] Omid Ardakanian et al. “On Identification of Distribution Grids”. In: *IEEE transactions on control of network systems* 6 (3 2019), pp. 950–960. ISSN: 2325-5870.

[35] Elena Gonzalez et al. “Using high-frequency SCADA data for wind turbine performance monitoring: A sensitivity study”. In: *Renewable Energy* 131 (2019), pp. 841–853. ISSN: 0960-1481. DOI: https://doi.org/10.1016/j.renene.2018.07.068. URL: https://www.sciencedirect.com/science/article/pii/S0960148118308656.

[36] Nordpool. *Day-ahead Prices*. Feb. 2026. URL: https://data.nordpoolgroup.com/auction/day-ahead/prices?deliveryDate=latest%C2%A4cy=SEK&aggregation=DeliveryPeriod&deliveryAreas=SE3.

[37] Reigh Walling et al. “Fault Current Contributions from Wind Plants”. In: *Proceedings of 2015 68th annual conference for protective relay engineers*. Mar. 2015, pp. 137–227.

[38] V Gevorgian and E Muljadi. “Wind Power Plant Short Circuit Current Contribution for Different Fault and Wind Turbine Topologies: Preprint”. In: *The 9th Annual International Workshop on Large-Scale Integration of Wind Power into Power Systems*. Oct. 2010.

[39] Jun Yin, Weichen Qian, and Xiaobo Huang. “A New Short-Circuit Current Calculation and Fault Analysis Method Suitable for Doubly Fed Wind Farm Groups”. In: *Sensors* 23 (20 2023). ISSN: 1424-8220. DOI: 10.3390/s23208372. URL: https://www.mdpi.com/1424-8220/23/20/8372.

[40] Zhongping Liu et al. “Local Iterative Calculation Method and Fault Analysis of Short-Circuit Current in High-Voltage Grid with Large-Scale New Energy Equipment Integration”. In: *Sustainability* 16 (24 2024). ISSN: 2071-1050. DOI: 10.3390/su162411144. URL: https://www.mdpi.com/2071-1050/16/24/11144.

[41] S Gao et al. “Equivalent Modeling of Three Phase Short Circuit Current Calculation of Direct-Drive Wind Farms Considering Internal Distribution Characteristics”. In: *2025 4th International Conference on Green Energy and Power Systems (ICGEPS)*. 2025, pp. 635–639. DOI: 10.1109/ICGEPS65133.2025.11034514.

[42] Yafang Tang, Guojuan Luo, and Yi Pu. “Short-circuit calculation approach of grids with high proportion of inverter-interfaced generator stations”. In: *Electric Power Systems Research* 241 (2025), p. 111407. ISSN: 0378-7796. DOI: https://doi.org/10.1016/j.epsr.2024.111407. URL: https://www.sciencedirect.com/science/article/pii/S0378779624012938.

[43] Po Li et al. “A Short-Circuit Current Calculation Model for Renewable Power Plants Considering Internal Topology”. In: *Applied Sciences* 14 (22 2024). ISSN: 2076-3417. DOI: 10.3390/app142210745. URL: https://www.mdpi.com/2076-3417/14/22/10745.

[44] M Fischer and A Mendonca. “Representation of variable speed full conversion Wind Energy Converters for steady state short-circuit calculations”. In: *2011 IEEE Power and Energy Society General Meeting*. IEEE, 2011, pp. 1–7. ISBN: 9781457710001.

[45] A Stiger, R A Rivas, and M Halonen. “Synchronous Condensers Contribution to Inertia and Short Circuit Current in Cooperation with STATCOM”. In: *2019 IEEE PES GTD Grand International Conference and Exposition Asia (GTD Asia)*. 2019, pp. 955–959. DOI: 10.1109/GTDAsia.2019.8715893.

[46] Chen Xing-wang et al. “Effective short circuit ratio including impact of STATCOM”. In: *2016 IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC)*. 2016, pp. 800–804. DOI: 10.1109/APPEEC.2016.7779605.

[47] Lian Li and Xiao Zhang. “Study on STATCOM principle and control strategy under short circuit fault”. In: *2017 IEEE International Conference on Mechatronics and Automation (ICMA)*. 2017, pp. 1187–1191. DOI: 10.1109/ICMA.2017.8015985.

[48] Aboutaleb Haddadi et al. “Need for Load Modeling in Short Circuit Analysis of an Inverter-Based Resource-Dominated Power System”. In: *IEEE Transactions on Power Delivery* 38 (3 2023), pp. 1882–1890. DOI: 10.1109/TPWRD.2022.3227138.

[49] Qianggang Wang, Niancheng Zhou, and Ling Ye. “Fault Analysis for Distribution Networks With Current-Controlled Three-Phase Inverter-Interfaced Distributed Generators”. In: *IEEE transactions on power delivery* 30 (3 2015), pp. 1532–1542. ISSN: 0885-8977.

[50] Hossein Hooshyar and Mesut E Baran. “Fault Analysis on Distribution Feeders With High Penetration of PV Systems”. In: *IEEE transactions on power systems* 28 (3 2013), pp. 2890–2896. ISSN: 0885-8950.

[51] Dao Van Tu, Surachai Chaitusaney, and Akihiko Yokoyama. “Maximum-Allowable Distributed Generation Considering Fault Ride-Through Requirement And Reach Reduction of Utility Relay”. In: *IEEE transactions on power delivery* 29 (2 2014), pp. 534–541. ISSN: 0885-8977.

[52] Felipe Proença de Albuquerque et al. “Comprehensive analysis of admittance matrix estimation considering different noise models”. In: *IET generation, transmission & distribution* 19 (1 2025). ISSN: 1751-8687.

[53] W H Kersting and Greg Shirek. “Short circuit analysis of IEEE test feeders”. In: *PES T&D 2012*. 2012, pp. 1–9. DOI: 10.1109/TDC.2012.6281539.

[54] Nasser D Tleis. *Power systems modelling and fault analysis : theory and practice*. 2nd ed. Academic Press, 2019. ISBN: 9780128151181.

[55] J C Das. *Short-circuits in AC and DC systems : AMSI, IEEE, and IEC standards*. 1st ed. Includes bibliographical references at the end of each chapters and index. CRC Press, Taylor & Francis Group, 2017. ISBN: 1-4987-4542-3. DOI: 10.1201/9781351228282.

[56] Paul M Anderson. “Computer Solution Methods Using the Impedance Matrix”. In: *Analysis of Faulted Power Systems*. Vol. 11. John Wiley & Sons, 2009. ISBN: 9780780311459.

[57] Electric Power Research Institute. *Ciruit Model Concept*. 2024. URL: https://opendss.epri.com/CircuitModelConcept.html.

[58] Paul M Anderson. “Sequence Impedance of Transformers”. In: *Analysis of Faulted Power Systems*. 1995, pp. 231–272. DOI: 10.1109/9780470544129.ch7.

[59] PowerWorld Corporation. *Load Modeling*. 2020. URL: https://www.powerworld.com/files/D05LoadModeling.pdf.

[60] Svenska Kraftnät. *Handbok kraftsystemmodellering*. Tech. rep. Oct. 2025.

[61] PowerWorld Corporation. *PowerWorld*. 2026. URL: https://www.powerworld.com.

[62] Leon Thurner et al. “pandapower - an Open Source Python Tool for Convenient Modeling, Analysis and Optimization of Electric Power Systems”. In: (Apr. 2018). DOI: 10.1109/TPWRS.2018.2829021.

[63] University of Washington. *14 Bus Power Flow Test Case*. Aug. 1993. URL: https://labs.ece.uw.edu/pstca/pf14/pg_tca14bus.htm.

[64] European Network of Transmission System Operators for Electricity. *Transparency platform*. URL: https://transparency.entsoe.eu/.

Appendix A

Methods parameters

Table A.1: Complete method parameters and those used in the reduced-order model. Tables in Reference as in Svenska Kraftnät modelling manual [60].

Element	Item	Description	Reduced order model	Reference
Transmission Line	inService	On/Off	x	Tabell 25
Transmission Line	r	Series resistance	x	Tabell 25
Transmission Line	x	Series reactance	x	Tabell 25
Transmission Line	g	Shunt conductance	x	Tabell 25
Transmission Line	b	Shunt susceptance	x	Tabell 25
Transformer	inService	On/Off	x	Tabell 29
Transformer	V_high	Voltage rating high	x	Tabell 30
Transformer	V_low	Voltage rating low	x	Tabell 30
Transformer	S	Nominal rating	x	Tabell 30
Transformer	r	Series resistance	x	Tabell 30
Transformer	x	Series reactance	x	Tabell 30
Transformer	m	Off-nominal tap ratio	x	
Uncontrolled Shunt	inService	On/Off	x	Tabell 51
Uncontrolled Shunt	V	Nominal voltage	x	Tabell 51
Uncontrolled Shunt	g	Shunt conductance	x	Tabell 52
Uncontrolled Shunt	b	Shunt susceptance	x	Tabell 52
Synchronous Generator	inService	On/Off	x	Tabell 42
Synchronous Generator	S	Nominal rating	x	Tabell 43
Synchronous Generator	V	Nominal voltage	x	Tabell 43
Synchronous Generator	r	Armature resistance	x	Tabell 44
Synchronous Generator	X”d	Subtransient reactance	x	Tabell 44
Synchronous Generator	V_meas	Voltage measurement		Tabell 58
Synchronous Generator	P_meas	Active power measurement		Tabell 58
Synchronous Generator	Q_meas	Reactive power measurement		Tabell 58
Full Converter	inService	On/Off		Tabell 42
Full Converter	S	Nominal rating		Tabell 42
Full Converter	cosPhi	Nominal power factor		Tabell 42
Full Converter	V	Nominal voltage		Tabell 2
Full Converter	K_scc	Maximum short circuit current		
Full Converter	V_min	Minimum voltage		*
Full Converter	K_factor	Controller K factor		
Full Converter	V_meas	Voltage measurement		Tabell 58
Full Converter	P_meas	Active power measurement		Tabell 58
Full Converter	Q_meas	Reactive power measurement		Tabell 58
External Grid	V	Nominal voltage	x	Tabell 44
External Grid	S_sc	Short circuit capacity	x	Tabell 44
Loads	inService	On/Off		Tabell 49
Loads	V	Nominal voltage		Tabell 2
Loads	P_rated	Nominal active power		Tabell 49
Loads	Q_rated	Nominal reactive power		Tabell 49
Loads	V_meas	Voltage measurement		Tabell 58
Loads	P_meas	Active power measurement		Tabell 58
Loads	Q_meas	Reactive power measurement		Tabell 58
Substation/Buses	V_meas	Voltage measurement		Tabell 58

* FRT defined in [19].